

22 Variational Calculus and Optimal Control

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Since the fabric of the universe is most perfect, and is the work of a most wise Creator, nothing whatsoever takes place in the universe in which some form of maximum and minimum does not appear.

Leonard Euler, 1744.

22.1 Basic lemmas of variation calculus

The following lemmas represent the basic instrument for proving the main results of *variation calculus theory and optimal control theory* (see, for example, Gel'fand & Fomin (1961), Ivanov & Faldin (1981) and Troutman (1996)).

22.1.1 Du Bois–Reymond lemma

First, let us prove the following simple auxiliary result.

Lemma 22.1. *If $0 \leq p \in C[a, b]$ and $\int_{x=a}^b p(x) dx = 0$, then $p(x) = 0$ for all $x \in [a, b]$.*

Proof. Since $p(x) \geq 0$, for any $x \in [a, b]$ we have

$$0 \leq P(x) := \int_{t=a}^x p(t) dt \leq \int_{t=a}^b p(t) dt = 0$$

So, $P(x) \equiv 0$ on $[a, b]$, and, hence, $P'(x) \equiv 0$ too. This exactly means $p(x) = P'(x) \equiv 0$ which proves the statement of this lemma. $\square$

Lemma 22.2. (Du Bois–Reymond) *If $h \in C[a, b]$ is a continuous on $[a, b]$ scalar function of the scalar argument and*

$$\int_{x=a}^b h(x) v'(x) dx = 0 \quad (22.1)$$

for all $v \in D_1 := \{v \in C^1[a, b] : v(a) = v(b) = 0\}$. Then

$$h(x) = c = \text{const on } [a, b]$$

Proof. For a constant c , the function $v(x) := \int_{t=a}^x [h(t) - c] dt$ is in $C^1[a, b]$ (it has a continuous derivative) and $v'(x) = h(x) - c$ so that $v(a) = 0$. It will be in D_1 if, additionally, it satisfies the condition $v(b) = 0$, that is, if $v(b) := \int_{t=a}^b [h(t) - c] dt = 0$, or $c = (b - a)^{-1} \int_{t=a}^b h(t) dt$. Thus, for these c and $v(x)$, in view of (22.1), we have

$$\begin{aligned} 0 &\leq \int_{x=a}^b [h(x) - c]^2 dx = \int_{x=a}^b [h(x) - c] v'(x) dx \\ &= \int_{x=a}^b h(x) v'(x) dx - c v(x) \Big|_{x=a}^{x=b} = 0 \end{aligned}$$

and, by Lemma 22.1, it follows that $[h(x) - c]^2 \equiv 0$ which completes the proof. $\square$

The next lemma generalizes Lemma 22.2.

Lemma 22.3. *If $h \in C[a, b]$ and for some $m = 1, 2, \dots$*

$$\int_{x=a}^b h(x) v^{(m)}(x) dx = 0$$

for all $v \in D_m$ where

$$D_m := \{v \in C^m[a, b] : v^{(k)}(a) = v^{(k)}(b) = 0, k = 0, 1, \dots, m-1\}$$

Then on $[a, b]$ the function $h(x)$ is a polynomial of a degree $l < m$, that is,

$$h(x) = c_0 + c_1 x + \dots + c_l x^l$$

Proof. By a translation, we may assume that $a = 0$. The function

$$H(x) := \int_{t_1=0}^x \left(\dots \left(\int_{t_{m-1}=0}^{t_{m-2}} \left(\int_{t=0}^{t_{m-1}} h(t) dt \right) dt_{m-1} \right) \dots \right) dt_1$$

is in $C^m[0, b]$ with the derivative $H^{(m)}(x) = h(x)$, and, fulfilling the identities

$$H(0) = H'(0) = \dots = H^{(m-1)}(0) = 0$$

Then, if q is a polynomial of a degree $l < m$, then $P(x) := x^m q(x)$ vanishes at $x = 0$ together with $P^{(i)}(x)$ for $i < m$, while $p(x) := P^{(m)}(x)$ is another polynomial of the degree $l < m$. Define $v(x) := H(x) - P(x)$, so that $v^{(m)}(x) = h(x) - p(x)$. Show next that with the proper choice of $q(x)$ we can make $v^{(k)}(b) = 0$ ($k = 0, 1, \dots, m-1$). Supposing that this choice has been made, the resulting $v \in D_m$, and, moreover,

$$\begin{aligned} \int_{x=0}^b p(x) v^{(m)}(x) dx &= - \int_{x=0}^b p'(x) v^{(m-1)}(x) dx \\ &= \dots = (-1)^m \int_{x=0}^b p^{(m)}(x) v(x) dx = 0 \end{aligned}$$

since the boundary term vanishes. By the assumptions of this lemma it follows that

$$\begin{aligned} 0 &\leq \int_{x=0}^b [h(x) - p(x)]^2 dx = \int_{x=0}^b [h(x) - p(x)] v^{(m)}(x) dx \\ &= \int_{x=0}^b h(x) v^{(m)}(x) dx = 0 \end{aligned}$$

So, by Lemma 22.1, we get $h(x) = p(x)$ on $[0, b]$. Lemma is proven. $\square$

Lemma 22.4. If $g, h \in C[a, b]$ and

$$\int_{x=a}^b [g(x) v(x) + h(x) v'(x)] dx = 0$$

for all $v \in D_1 := \{v \in C^1[a, b] : v(a) = v(b) = 0\}$. Then $h \in C^1[a, b]$ and

$$h'(x) = g(x) \quad \text{for all } x \in [a, b]$$

Proof. Denote $G(x) := \int_{t=a}^x g(t) dt$ for $x \in [a, b]$. Then $G \in C^1[a, b]$ and $G'(x) = g(x)$. The integration by part implies

$$\begin{aligned} 0 &= \int_{x=a}^b [g(x) v(x) + h(x) v'(x)] dx = \int_{x=a}^b [h(x) - G(x)] v'(x) dx \\ &\quad + G'(x) v(x) \Big|_{x=a}^{x=b} = \int_{x=a}^b [h(x) - G(x)] v'(x) dx \end{aligned}$$

By Lemma 22.2, it follows that $h(x) - G(x) \equiv c = \text{const}$, so that

$$h(x) = G(x) + c, \quad h'(x) = G'(x) = g(x)$$

which proves the lemma. $\square$

Corollary 22.1. *If $g \in C[a, b]$ and $\int_{x=a}^b g(x) v(x) dx = 0$ for all $v \in D_1$, then $g(x) \equiv 0$ on $[a, b]$.*

Proof. It is sufficient to put $h(x) \equiv 0$ in Lemma 22.4. $\square$

This corollary also admits the following generalization.

22.1.2 Lagrange lemma

Lemma 22.5. (Lagrange) *If $g \in C[a, b]$ and for some $m = 0, 1, \dots$*

$$\int_{x=a}^b g(x) v(x) dx = 0$$

for all $v \in D_m$ where

$$D_m := \{v \in C^m[a, b] : v^{(k)}(a) = v^{(k)}(b) = 0, \quad k = 0, 1, \dots, m-1\}$$

then $g(x) \equiv 0$ on $[a, b]$.

Proof. Suppose, by contradiction, that $g(c) > 0$ for some $c \in (a, b)$. Then by continuity, there exists an interval $[\alpha, \beta] \subseteq (a, b)$ which contains c and such that

$$|g(x) - g(c)| \leq g(c)/2$$

or, equivalently, $g(x) \geq g(c)/2 > 0$. On the other hand, the function

$$v(x) := \begin{cases} (x-a)(\beta-x)^{m+1} & \text{for } x \in [\alpha, \beta] \\ 0 & \text{for } x \notin [\alpha, \beta] \end{cases}$$

is in $C^m[a, b]$ and nonnegative. This implies that the product $g(x) v(x)$ is also continuous, nonnegative, and not identically zero. Thus, $\int_{x=a}^b g(x) v(x) dx > 0$, which contradicts the hypotheses of this lemma. Lemma is proven. $\square$

The vector-valued version of Lemma 22.4 is also admitted.

Lemma 22.6. *If $d = 1, 2, 3, \dots$ and $G, H \in (C[a, b])^n$ ($G(x), H(x) \in \mathbb{R}^n$) so that*

$$\int_{x=a}^b [(G(x), H(x)) + (H(x), V'(x))] dx = 0$$

for any $V \in D_1 := \{v \in (C^1[a, b])^n : V(a) = V(b) = \bar{0}\}$ (here $\bar{0}$ is the zero-vector), then $H \in (C^1[a, b])^n$ and $H'(x) = G(x)$ on $[a, b]$.

Proof. It follows directly from Lemma 22.4 after its application to each individual component taking $V(x) = \left(\underbrace{0, \dots, v(x)}_i, 0, \dots, 0 \right)^T$ for $i = 1, \dots, n$. $\square$

Corollary 22.2. If $H \in (C[a, b])^n$ and $\int_{x=a}^b (G(x), H(x)) dx = 0$ for all $V \in D_1$, then $H(x) = \text{const} \in \mathbb{R}^n$ on $[a, b]$.

Proof. It is sufficient to take $G(x) \equiv 0$ in Lemma 22.6. $\square$

22.1.3 Lemma on quadratic functionals

Lemma 22.7. (Gel'fand & Fomin 1961) If $q, p \in C[a, b]$ and

$$I(y) = \int_{x=a}^b \left[q(x) y^2(x) + p(x) (y'(x))^2 \right] dx \geq 0 \quad (22.2)$$

for any function $y \in D_1 := \{y \in C^1[a, b] : y(a) = y(b) = 0\}$, then $p(x) \geq 0$ on $[a, b]$.

Proof. By the contradiction method, suppose that there exists $x_0 \in (a, b)$ such that $p(x_0) < 0$. Select the function

$$y(x) = y_\sigma(x) := \begin{cases} \sqrt{\sigma} \left(1 + \frac{x - x_0}{\sigma} \right) & \text{if } x_0 - \sigma \leq x \leq x_0 \\ \sqrt{\sigma} \left(1 - \frac{x - x_0}{\sigma} \right) & \text{if } x_0 \leq x \leq x_0 + \sigma \\ 0 & \text{if } x \notin [x_0 - \sigma, x_0 + \sigma] \end{cases}$$

(here $\sigma > 0$ is small enough). Then $(y'_\sigma(x))^2 = \sigma^{-1}$ and, by the mean-value theorem,

$$\left| \int_{x=a}^b q(x) y_\sigma^2(x) dx \right| = \left| \int_{x=x_0-\sigma}^{x_0+\sigma} q(x) y_\sigma^2(x) dx \right| \leq |q(x_0)| 2\sigma^2$$

and

$$\int_{x=a}^b p(x) (y'_\sigma(x))^2 dx = \int_{x=x_0-\sigma}^{x_0+\sigma} p(x) (y'_\sigma(x))^2 dx = 2p(\bar{x})$$

where $\bar{x} \in [x_0 - \sigma, x_0 + \sigma]$. For small enough σ , so that $p(\bar{x}) < 0$, we have

$$\begin{aligned} I(y) &\leq \left| \int_{x=a}^b q(x) y_\sigma^2(x) dx \right| + \int_{x=a}^b p(x) (y'_\sigma(x))^2 dx \\ &\leq |q(x_0)| 2\sigma^2 + 2p(\bar{x}) < 0 \end{aligned}$$

which contradicts the hypotheses of this lemma. So, $p(x_0) \geq 0$ for any internal point of the interval $[a, b]$. As for the boundary points a, b , the values $p(a), p(b)$ also cannot be negative since, if so, by the continuity property, $p(x)$ should be negative in a small internal neighborhood which is impossible. Lemma is proven. $\square$

22.2 Functionals and their variations

Here we will briefly remember the main definitions of the first and second variation of functionals in some functional Banach space $\mathcal{B}$.

Definition 22.1. The functional $J(y)$ defined in a Banach space $\mathcal{B}$ with a norm $\|\cdot\|_{\mathcal{B}}$ is said to be

1. **strongly differentiable** (in the **Fréchet sense**) at the “point” $y_0 \in \mathcal{B}$, if there exists a linear (with respect to variation $h \in \mathcal{B}$) functional $\varphi_1(y_0, h)$ such that for any $h \in \mathcal{B}$

$$\begin{aligned} \Delta J(y_0, h) &:= J(y_0 + h) - J(y_0) \\ &= \varphi_1(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}} \end{aligned} \tag{22.3}$$

where $\alpha(y_0, h) \rightarrow 0$ as $\|h\|_{\mathcal{B}} \rightarrow 0$;

- 2 **twice strongly differentiable** (in the **Fréchet sense**) at the “point” $y_0 \in \mathcal{B}$, if $\Delta J(y_0, h)$ (22.3) can be represented as

$$\Delta J(y_0, h) = \varphi_1(y_0, h) + \frac{1}{2} \varphi_2(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}}^2 \tag{22.4}$$

where $\varphi_1(y_0, h)$ is a **linear** (with respect to the variation $h \in \mathcal{B}$) functional, $\varphi_2(y_0, h)$ is a **quadratic functional** with respect to the variation $h \in \mathcal{B}$, i.e., for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and any $h, h_1, h_2, \tilde{h}, \tilde{h}_1, \tilde{h}_2 \in \mathcal{B}$

$$\begin{aligned} \varphi_2(y_0, h) &:= \tilde{\varphi}_2(y_0, h, \tilde{h})|_{\tilde{h}=h \in \mathcal{B}} \\ \tilde{\varphi}_2(y_0, \lambda_1 h_1 + \lambda_2 h_2, \tilde{h}) &= \lambda_1 \tilde{\varphi}_2(y_0, h_1, \tilde{h}) + \lambda_2 \tilde{\varphi}_2(y_0, h_2, \tilde{h}) \\ \tilde{\varphi}_2(y_0, h, \lambda_1 \tilde{h}_1 + \lambda_2 \tilde{h}_2) &= \lambda_1 \tilde{\varphi}_2(y_0, h, \tilde{h}_1) + \lambda_2 \tilde{\varphi}_2(y_0, h, \tilde{h}_2) \end{aligned}$$

and, again, $\alpha(y_0, h) \rightarrow 0$ as $\|h\|_{\mathcal{B}} \rightarrow 0$.

Definition 22.2. The functionals $\varphi_1(y_0, h)$ and $\varphi_2(y_0, h)$, defined above, are called the **first and second strong differentials** of $J(y)$ and are denoted (according to (18.93)) by

$$\begin{aligned} d\Phi(x_0 | h) &= \langle h, \Phi'(x_0) \rangle := \varphi_1(y_0, h) \\ d^2\Phi(x_0 | h) &:= \varphi_2(y_0, h) \end{aligned} \quad (22.5)$$

There exist other differentials, namely, weak differentials (in the Gâteaux sense). For details concerning these functionals and their relation with strong ones see section 18.7.2. Below we shall use only Fréchet differentials.

22.3 Extremum conditions

22.3.1 Extremal curves

The, so-called, *variation principle* (see Theorem 18.18) will be actively used in this section for the solution of various problems of variation calculus theory.

Definition 22.3.

1. A functional $J(y)$, defined in a Banach space $\mathcal{B}$ with a norm $\|\cdot\|_{\mathcal{B}}$, has a **local extremum in a region** $\mathcal{G}$ (defining some additional constraints to the class of admissible functions) at the curve (function) $y_0 \in \mathcal{B} \cap \mathcal{G}$, if there exists a neighborhood

$$\Omega_\delta := \{y \in \mathcal{B} \cap \mathcal{G} \mid \|y - y_0\|_{\mathcal{B}} < \delta\}$$

such that for all $y \in \Omega_\delta$ one has $J(y) \geq J(y_0)$. The function y_0 is said to be an **extremal curve**.

2. If $J(y) \geq J(y_0)$ for all $y \in \mathcal{B} \cap \mathcal{G}$, then the extremal curve y_0 is said to be a **global extremum** of the functional $J(y)$ on $\mathcal{B} \cap \mathcal{G}$.

22.3.2 Necessary conditions

Reformulate here Theorem 18.18 for the case of the Fréchet differential existence.

Theorem 22.1. (on the necessary conditions) Let the curve $y_0 \in \text{int}(\mathcal{B} \cap \mathcal{G})$ be a local extremal (minimal) curve of the functional $J(y)$ which is assumed to be strongly (Fréchet) differentiable at the “point” y_0 .

1. (**The first-type necessary conditions**) Then for any admissible $h \in \mathcal{B} \cap \mathcal{G}$

$$\boxed{\varphi_1(y_0, h) \equiv 0} \quad (22.6)$$

2. (**The second-type necessary conditions**) If, additionally, $J(y)$ is twice strongly (Fréchet) differentiable at the “point” y_0 , then for any admissible $h \in \mathcal{B} \cap \mathcal{G}$

$$\boxed{\varphi_2(y_0, h) \geq 0} \quad (22.7)$$

Proof. Let $J(y) \geq J(y_0)$ for all y within some Ω_δ .

1. Then, by Definition 22.3 and in view of the property (22.3), for any $(y_0 + h) \in \Omega_\delta$ we have

$$\Delta J(y_0, h) := J(y_0 + h) - J(y_0) = \varphi_1(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}} \geq 0$$

If $\varphi_1(y_0, h_0) < 0$ for some admissible h_0 , then for small enough α it follows that

$$\Delta J(y_0, h) = \varphi_1(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}} < 0$$

which contradicts the optimality of y_0 in Ω_δ . Suppose that $\varphi_1(y_0, h) > 0$ for some admissible h_0 . Since the strong (Fréchet) differential is linear on h and $y_0 \in \text{int}(\mathcal{B} \cap \mathcal{G})$, it follows that $(y_0 - h_0) \in \Omega_\delta$ and, therefore,

$$\varphi_1(y_0, -h_0) = -\varphi_1(y_0, h_0) < 0$$

and, as the result, again

$$\Delta J(y_0, h) = -\varphi_1(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}} < 0$$

for small enough α , which contradicts the optimality of y_0 in Ω_δ .

2. According to the first-type necessary condition (22.6) in view of the optimality of the curve $y_0 \in \text{int}(\mathcal{B} \cap \mathcal{G})$ we have

$$\Delta J(y_0, h) = \frac{1}{2} \varphi_2(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}}^2 \geq 0$$

Again, if for some admissible h_0 ($(y_0 - h_0) \in \Omega_\delta$) we suppose $\varphi_2(y_0, h_0) < 0$, then for small enough $\varepsilon > 0$ we have

$$\begin{aligned} \Delta J(y_0, \varepsilon h_0) &= \frac{1}{2} \varphi_2(y_0, \varepsilon h_0) + \alpha(y_0, \varepsilon h_0) \|\varepsilon h_0\|_{\mathcal{B}}^2 \\ &= \varepsilon^2 \left[\frac{1}{2} \varphi_2(y_0, h_0) + \alpha(y_0, \varepsilon h_0) \|h_0\|_{\mathcal{B}}^2 \right] < 0 \end{aligned}$$

since $\alpha(y_0, \varepsilon h_0) \rightarrow 0$ whereas $\varepsilon \rightarrow 0$ for any h_0 . But the last inequality contradicts the condition of optimality of the curve y_0 . $\square$

22.3.3 Sufficient conditions

Theorem 22.2. (on the sufficient conditions) *Let*

1. *the functional $J(y)$ be twice strongly (Fréchet) differentiable in $\mathcal{B} \cap \mathcal{G}$;*
2. *for some $y_0 \in \mathcal{B} \cap \mathcal{G}$ and any admissible $h : (y_0 - h) \in \mathcal{B} \cap \mathcal{G}$, $h \in \Omega_\delta$*

$\varphi_1(y_0, h) \equiv 0$

and

$$\varphi_2(y_0, h) \geq k \|h\|_{\mathcal{B}}^2, \quad k > 0$$

Then y_0 is the unique local minimal curve of the functional $J(y)$ on $\mathcal{B} \cap \mathcal{G}$.

Proof.

1. By (22.4) and in view of the condition of this theorem, for any admissible h

$$\Delta J(y_0, h) = \frac{1}{2} \varphi_2(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}}^2$$

For any small enough $\varepsilon > 0$ it follows that

$$\begin{aligned} \Delta J(y_0, \varepsilon h) &= \frac{1}{2} \varphi_2(y_0, \varepsilon h) + \alpha(y_0, \varepsilon h) \|\varepsilon h\|_{\mathcal{B}}^2 \\ &= \varepsilon^2 \left[\frac{1}{2} \varphi_2(y_0, h) + \alpha(y_0, \varepsilon h) \|h\|_{\mathcal{B}}^2 \right] \\ &\geq \varepsilon^2 \left[\frac{1}{2} k \|h\|_{\mathcal{B}}^2 + \alpha(y_0, \varepsilon h) \|h\|_{\mathcal{B}}^2 \right] \\ &= \varepsilon^2 \|h\|_{\mathcal{B}}^2 \left[\frac{1}{2} k + \alpha(y_0, \varepsilon h) \right] \geq 0 \end{aligned} \tag{22.8}$$

since $\alpha(y_0, \varepsilon h) \rightarrow 0$ whereas $\varepsilon \rightarrow 0$ for any admissible h . This means that y_0 is a local extremal (minimum) curve.

2. Suppose that in Ω_δ there exists two extremal curves y_0 and y'_0 such that $J(y_0) = J(y'_0) \leq J(y)$. Then, taking $h := y'_0 - y_0$ in (22.8), we get

$$\begin{aligned} \Delta J(y_0, h) = 0 &= \frac{1}{2} \varphi_2(y_0, h) + \alpha(y_0, h) \|h\|_{\mathcal{B}}^2 \\ &\geq \|h\|_{\mathcal{B}}^2 \left[\frac{1}{2} k + \alpha(y_0, h) \right] \end{aligned}$$

For small enough δ and any $h \in \Omega_\delta$, we have $\frac{1}{2} k + \alpha(y_0, h) > 0$ which together with the previous inequality leads to the following conclusions: $0 = \|h\|_{\mathcal{B}}^2$, or, equivalently, $y'_0 = y_0$. Theorem is proven. $\square$

22.4 Optimization of integral functionals

In this section we will consider the following *main problem* of variation calculus:

$$\int_{x=a}^b F(x, y, y') dx \rightarrow \min_{y \in C^1[a, b]} \tag{22.9}$$

where the function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ is assumed to be *twice differentiable* in all arguments.

22.4.1 Curves with fixed boundary points

22.4.1.1 Scalar case

Here, we will additionally suppose that we are looking for the extremum (minimum) of the integral function in (22.9) over all continuously differentiable curves $y \in C^1[a, b]$ satisfying the following boundary conditions:

$$\boxed{y(a) = \alpha, \quad y(b) = \beta} \quad (22.10)$$

The necessary conditions for this problem are given below.

1. *The first-type necessary condition (22.6):*

$$\begin{aligned} 0 = \varphi_1(y, h) &= \int_{x=a}^b \left(\frac{\partial}{\partial y} F h + \frac{\partial}{\partial y'} F h' \right) dx \\ &= \int_{x=a}^b \left(\frac{\partial}{\partial y} F - \frac{d}{dx} \frac{\partial}{\partial y'} F \right) h dx \end{aligned} \quad (22.11)$$

where the variation curves $h \in C^1[a, b]$ and satisfies the boundary conditions $h(a) = h(b)$.

2. *The second-type necessary condition (22.7):*

$$\begin{aligned} 0 &\leq \varphi_2(y, h) \\ &= \int_{x=a}^b \left[\frac{\partial^2}{\partial y^2} F h^2 + 2 \frac{\partial^2}{\partial y \partial y'} F h h' + \frac{\partial^2}{\partial y' \partial y'} F (h')^2 \right] dx \\ &= \int_{x=a}^b \left[\left(\frac{\partial^2}{\partial y^2} F - \frac{d}{dx} \frac{\partial^2}{\partial y \partial y'} F \right) h^2 + \frac{\partial^2}{\partial y' \partial y'} F (h')^2 \right] dx \end{aligned} \quad (22.12)$$

where $y(x)$ satisfies (22.11) and the variation curves $h \in C^1[a, b]$ and fulfills the boundary conditions $h(a) = h(b)$.

In (22.11) and (22.12) the following relations obtained by the integration by parts are used:

$$\begin{aligned} \int_{x=a}^b \frac{\partial}{\partial y'} F h' dx &= \left(\frac{\partial}{\partial y'} F \right) h \Big|_{x=a}^{x=b} - \int_{x=a}^b \left(\frac{d}{dx} \frac{\partial}{\partial y'} F \right) h dx \\ &= \int_{x=a}^b \left(-\frac{d}{dx} \frac{\partial}{\partial y'} F \right) h dx \end{aligned}$$

and

$$\begin{aligned} \int_{x=a}^b 2 \frac{\partial^2}{\partial y \partial y'} F h h' dx &= \left(\frac{\partial^2}{\partial y \partial y'} F \right) h^2 \Big|_{x=a}^{x=b} - \int_{x=a}^b \left(\frac{d}{dx} \frac{\partial^2}{\partial y \partial y'} F \right) h^2 dx \\ &= \int_{x=a}^b \left(- \frac{d}{dx} \frac{\partial^2}{\partial y \partial y'} F \right) h^2 dx \end{aligned}$$

Theorem 22.3. (Euler–Lagrange) *The first-type necessary condition for a curve $y \in C^1[a, b]$, satisfying (22.10), to be an extremal curve is:*

$$\boxed{\frac{\partial}{\partial y} F(x, y, y') - \frac{d}{dx} \frac{\partial}{\partial y'} F(x, y, y') = 0} \quad (22.13)$$

Proof. It follows directly from (22.11) if we apply Lemma 22.4. $\square$

The condition (22.13) is referred to as the *Euler–Lagrange condition*.

Theorem 22.4. (Legendre) *The second-type necessary condition for a curve $y \in C^1[a, b]$, satisfying (22.10) and (22.13), to be an extremal (minimizing) curve is:*

$$\boxed{\frac{\partial^2}{\partial y' \partial y'} F(x, y, y') \geq 0} \quad (22.14)$$

Proof. It follows directly from (22.12) if we apply Lemma 22.7. $\square$

The condition (22.14) is often referred to as *the Legendre condition*.

The *sufficient conditions*, guaranteeing that a curve $y(x)$ is minimizing for the functional $J(y)$, may be formulated in the following manner.

Theorem 22.5. (Jacobi) *If for some curve $y \in C^1[a, b]$, verifying (22.10), the following conditions are fulfilled:*

1. *it satisfies the **Euler–Lagrange necessary condition** (22.13);*
2. *it satisfies the **strong Legendre condition***

$$\boxed{\frac{\partial^2}{\partial y' \partial y'} F(x, y, y') \geq k > 0} \quad (22.15)$$

3. *there exists a function $u \in C^1[a, b]$, which is not equal to zero on $[a, b]$, and satisfying the next ODE (the **Jacobi ODE**):*

$$\boxed{\begin{aligned} Qu - \frac{d}{dx} (Pu') &= 0 \\ Q &:= \frac{\partial^2}{\partial y^2} F - \frac{d}{dx} \frac{\partial^2}{\partial y \partial y'} F, \quad P := \frac{\partial^2}{\partial y' \partial y'} F \end{aligned}} \quad (22.16)$$

then this curve provides the local minimum to the functional $J(y)$.

Proof. It follows directly from Theorem 22.2 if we define $\|h\|_{\mathcal{B}}^2$ for $\mathcal{B} = C^1[a, b]$ (see (18.6)) as

$$\|h\|_{C^1[a, b]} := \max_{x \in [a, b]} |h(x)| + \max_{x \in [a, b]} |h'(x)|$$

Indeed, by the assumptions of this theorem, we have

$$\begin{aligned} \varphi_2(y, h) &= \int_{x=a}^b \left[Qh^2 + P(h')^2 \right] dx = \int_{x=a}^b \left[(Qu) \frac{h^2}{u} + P(h')^2 \right] dx \\ &= \int_{x=a}^b \left[\frac{d}{dx} (Pu') \frac{h^2}{u} + P(h')^2 \right] dx = \left(Pu' \frac{h^2}{u} \right) \Big|_{x=a}^{x=b} \\ &\quad + \int_{x=a}^b \left[-Pu' \frac{d}{dx} \left(\frac{h^2}{u} \right) + P(h')^2 \right] dx = \int_{x=a}^b P \left[(h')^2 - u' \frac{d}{dx} \left(\frac{h^2}{u} \right) \right] dx \\ &= \int_{x=a}^b P \left[(h')^2 - 2h' \left(\frac{hu'}{u} \right) + \frac{h^2}{u^2} (u')^2 \right] dx = \int_{x=a}^b P \left(h' - \frac{hu'}{u} \right)^2 dx \\ &\geq k \int_{x=a}^b \left(h' - \frac{hu'}{u} \right)^2 dx = k \|h\|_{C^1[a, b]}^2 \int_{x=a}^b \left(\tilde{h}' - \frac{\tilde{h}u'}{u} \right)^2 dx \end{aligned}$$

where $\tilde{h} := h/\|h\|_{C^1[a, b]}$ satisfies

$$\|\tilde{h}\|_{C^1[a, b]} \leq 1, \quad \|\tilde{h}'\|_{C^1[a, b]} \leq 1$$

Notice also that

$$\varkappa := \inf_{\tilde{h}: \|\tilde{h}\|_{C^1[a, b]} \leq 1, \|\tilde{h}'\|_{C^1[a, b]} \leq 1, \tilde{h}(a)=\tilde{h}(b)=0} \int_{x=a}^b \left(\tilde{h}' - \frac{\tilde{h}u'}{u} \right)^2 dx > 0$$

since, if not, one has $\tilde{h}' - \frac{\tilde{h}u'}{u} = 0$, and, as a result, $\tilde{h} = u$. But, by the assumption of this theorem, $u(a) \neq 0$ and $u(b) \neq 0$, which contradicts with $\tilde{h}(a) = \tilde{h}(b) = 0$. So, finally, it follows that

$$\varphi_2(y, h) \geq k \|h\|_{C^1[a, b]}^2 \int_{x=a}^b \left(\tilde{h}' - \frac{\tilde{h}u'}{u} \right)^2 dx \geq k \varkappa \|h\|_{C^1[a, b]}^2$$

Theorem is proven. $\square$

Example 22.1. (“Brachistochrone” problem) The problem (formulated by Johann Bernoulli, 1696) consists of finding the curve AB (see Fig. 22.1) such that, during the sliding over of this curve in a gravity field with the initial velocity equal to zero, the

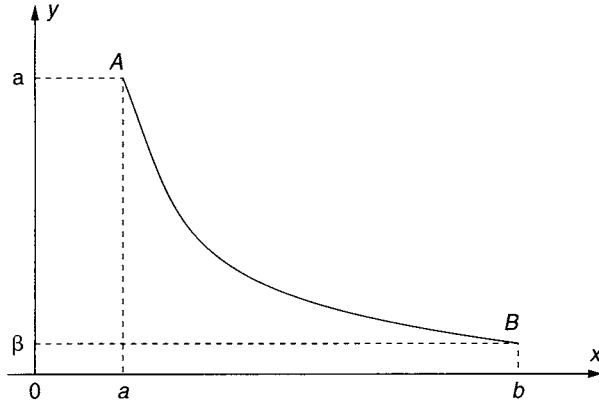


Fig. 22.1. Illustration of the Brachistochrone problem.

material point of the mass m can realize the sliding from the initial point A to the final point B in the shortest time.

So, we have to minimize $J(y) = T(y)$ where $y(x)$ is the altitude of the point at the curve at the point $x \in [a, b]$ which satisfies the relation

$$mgy + \frac{mv^2}{2} = E = \text{const}$$

Since, by the initial condition, when $y = y(a) = \alpha$ we have $v = 0$ which implies $E = mg\alpha$. That is why

$$v = \sqrt{2 \left(\frac{E}{m} - gy \right)} = \sqrt{2g(\alpha - y)}$$

$$v = \frac{ds}{dt} = \sqrt{1 + (y')^2} \frac{dx}{dt}$$

which leads to the following formula for the functional $T(y)$:

$$T(y) = \int_{x=a}^b \frac{dx}{v} = \frac{1}{\sqrt{2g}} \int_{x=a}^b \sqrt{\frac{1 + (y')^2}{(\alpha - y)}} dx$$

Since the function $F(y, y') = \sqrt{\frac{1 + (y')^2}{(\alpha - y)}}$ does not depend on x , the first-type Euler–Lagrange condition (22.13) for this functional has the **first integral**

$$F(y, y') - y' \frac{\partial}{\partial y'} F(y, y') = \frac{1}{c} \quad (22.17)$$

Indeed,

$$0 = \frac{\partial}{\partial y} F - \frac{d}{dx} \frac{\partial}{\partial y'} F = \frac{\partial}{\partial y} F - \left(\frac{\partial^2}{\partial y \partial y'} F \right) y' - \left(\frac{\partial^2}{\partial y'^2 \partial y'} F \right) y''$$

Multiplying both sides by y' we get

$$0 = \frac{d}{dx} \left(F - y' \frac{\partial}{\partial y'} F \right)$$

which implies (22.17). In our case it is equivalent to the following ODE:

$$\begin{aligned} \frac{1}{c} &= \sqrt{\frac{1 + (y')^2}{(\alpha - y)}} - (y')^2 \frac{(1 + (y')^2)^{-1/2}}{\sqrt{(\alpha - y)}} \\ &= \frac{\sqrt{1 + (y')^2} \left(1 - \frac{(y')^2}{1 + (y')^2} \right)}{\sqrt{(\alpha - y)}} = \frac{1}{\sqrt{(\alpha - y)} \sqrt{1 + (y')^2}} \end{aligned}$$

Squaring gives

$$c^2 = (\alpha - y) \left[1 + (y')^2 \right]$$

or,

$$1 = y' \sqrt{\frac{(\alpha - y)}{c^2 - (\alpha - y)}} \quad (22.18)$$

With the introduction of the dependent variable $\theta = \theta(x)$ such that

$$(\alpha - y) = c^2 \sin^2 \frac{\theta}{2} = \frac{c^2}{2} (1 - \cos \theta), \quad \theta \in [0, 2\pi]$$

then

$$c^2 - (\alpha - y) = c^2 \cos^2 \frac{\theta}{2}, \quad y' = c^2 \theta' \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

By substitution of these expressions into (22.18) yields

$$1 = c^2 \theta' \sin^2 \frac{\theta}{2} \quad \text{or} \quad 1 = \frac{c^2}{2} \theta' (1 - \cos \theta)$$

Integrating gives $\frac{c^2}{2} [\theta - \sin \theta] = x - c_1$. Denoting $c_2 := \frac{c^2}{2}$, we get the **parametric (Brachistochrone) curve** (of the cycloid type)

$$\boxed{\begin{aligned} x &= c_2 [\theta - \sin \theta] + c_1, \quad c_2 > 0 \\ y &= \alpha - c_2 (1 - \cos \theta), \quad \theta \in [0, 2\pi] \end{aligned}} \quad (22.19)$$

The constants c_1 and c_2 can be found from the boundary conditions

$$\begin{aligned} \theta &= \theta_A = 0 : A = c_1, \quad y(A) = \alpha \\ \theta &= \theta_B : B = A + c_2 [\theta_B - \sin \theta_B], \quad y(B) = \beta = \alpha - c_2 (1 - \cos \theta_B) \end{aligned}$$

Example 22.2. Let us try to answer the following question: which stable linear system of the first order, given by

$$\dot{x}(t) = ax(t), \quad x(0) = x_0, \quad x(\infty) = 0 \quad (22.20)$$

provides the minimum of the functional

$$J(x) := \int_{t=0}^{\infty} [\alpha_0 x^2(t) + \alpha_1 \dot{x}^2(t)] dt \quad (22.21)$$

where $\alpha_0, \alpha_1 > 0$.

1. Variation calculus application. The first-type necessary condition of the optimality for this functional is

$$\alpha_0 x(t) - \alpha_1 \ddot{x}(t) = 0$$

since $F = \alpha_0 x^2 + \alpha_1 \dot{x}^2$. Its general solution is

$$x(t) = c_1 e^{-kt} + c_2 e^{kt}, \quad k := \sqrt{\alpha_0/\alpha_1}$$

Taking into account the boundary conditions we get: $c_2 = 0$, $c_1 = x_0$ which gives $x(t) = x_0 e^{-kt}$, or, equivalently, $\dot{x}(t) = -kx(t)$. So, the optimal $a = a^*$ in (22.20) is $a = a^* = -\sqrt{\alpha_0/\alpha_1}$. Let us show that the obtained curve is minimizing. To do this we need to check conditions (2) and (3) of the Jacobi theorem 22.5. First, notice that $\frac{\partial^2}{\partial \dot{x} \partial \dot{x}} F = 2\alpha_1 > 0$. So, condition (2) is fulfilled. The Jacobi equation (22.16) is as follows

$$\alpha_0 u - \alpha_1 u'' = 0$$

$$Q := \frac{\partial^2}{\partial y^2} F - \frac{d}{dx} \frac{\partial^2}{\partial y \partial y'} F = 2\alpha_0, \quad P := \frac{\partial^2}{\partial y' \partial y'} F = 2\alpha_1$$

It has a nontrivial solution $u(t) = u_0 e^{-kt} > 0$ or $u_0 > 0$. This means that condition (3) is also valid. So, the curve $\dot{x}(t) = -kx(t)$, $x(0) = x_0$ is minimizing.

2. Direct method.¹ Assuming that the minimal value of the cost functional (22.21) is finite, i.e., $x(t) \rightarrow 0$ with $t \rightarrow \infty$, we can represent it in the following equivalent form

$$\begin{aligned} J(x) &:= \int_{t=0}^{\infty} [\alpha_0 x^2(t) + 2\sqrt{\alpha_0 \alpha_1} x(t) \dot{x}(t) + \alpha_1 \dot{x}^2(t)] dt \\ &\quad - 2\sqrt{\alpha_0 \alpha_1} \int_{t=0}^{\infty} x(t) \dot{x}(t) dt = \int_{t=0}^{\infty} [\sqrt{\alpha_0} x(t) + \sqrt{\alpha_1} \dot{x}(t)]^2 dt \\ &\quad + \sqrt{\alpha_0 \alpha_1} x^2(0) \geq \sqrt{\alpha_0 \alpha_1} x^2(0) \end{aligned}$$

¹ The author has been informed of this elegant and simple solution by Prof. V. Utkin.

The last inequality becomes the equality when

$$\sqrt{\alpha_0}x(t) + \sqrt{\alpha_1}\dot{x}(t) = 0$$

or, equivalently, when

$$\dot{x}(t) = -kx(t), \quad k = \sqrt{\alpha_0/\alpha_1}$$

which coincides with the previous result.

22.4.1.2 Vector case

Consider the following optimization problem

$$\int_{x=a}^b F(x, y_1, \dots, y_n, y'_1, \dots, y'_n) dx \rightarrow \min_{y_i \in C^1[a, b]} \quad (22.22)$$

where the function $F : \mathbb{R}^{2n+1} \rightarrow \mathbb{R}$ is assumed to be *twice differentiable* in all arguments, and the functions y_i satisfy the following boundary conditions

$$y_i(a) = \alpha_i, \quad y_i(b) = \beta_i \quad (i = 1, \dots, n) \quad (22.23)$$

Theorem 22.6. (The Euler–Lagrange vector form) *The first-type necessary condition for curve $y_i \in C^1[a, b]$, satisfying (22.23), to be extremal curves is:*

$$\begin{aligned} \frac{\partial}{\partial y_i} F(x, \mathbf{y}, \mathbf{y}') - \frac{d}{dx} \frac{\partial}{\partial y'_i} F(x, \mathbf{y}, \mathbf{y}') &= 0 \\ \mathbf{y} &:= (y_1, \dots, y_n)^\top, \quad \mathbf{y}' := (y'_1, \dots, y'_n)^\top \end{aligned} \quad (22.24)$$

Proof. It follows directly from the identity

$$\begin{aligned} 0 = \varphi_1(y, h) &= \int_{x=a}^b \sum_{i=1}^n \left(\frac{\partial}{\partial y_i} F h_i + \frac{\partial}{\partial y'_i} F h'_i \right) dx \\ &= \int_{x=a}^b \sum_{i=1}^n \left(\frac{\partial}{\partial y_i} F_i - \frac{d}{dx} \frac{\partial}{\partial y'_i} F \right) h_i dx \end{aligned}$$

if we take into account the independence of the variation functions h_i ($i = 1, \dots, n$), and apply Lemma 22.4. $\square$

Theorem 22.7. (The Legendre vector form) *The second-type necessary condition for curve $y_i \in C^1[a, b]$, satisfying (22.23) and (22.24), to be extremal curves is:*

$$\nabla_{\mathbf{y}', \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \geq 0 \quad (22.25)$$

where the matrix in (22.25) is defined by

$$\nabla_{\mathbf{y}, \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') := \left[\frac{\partial^2}{\partial y'_i \partial y'_j} F(x, \mathbf{y}, \mathbf{y}') \right]_{i, j=1, \dots, n}$$

Proof. It follows directly from (22.7) and the relation

$$\begin{aligned} 0 \leq \varphi_2(y, h) &= \int_{x=a}^b [(\nabla_{\mathbf{y}, \mathbf{y}}^2 F(x, \mathbf{y}, \mathbf{y}') \mathbf{h}, \mathbf{h}) \\ &\quad + 2(\nabla_{\mathbf{y}, \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \mathbf{h}, \mathbf{h}') + (\nabla_{\mathbf{y}', \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \mathbf{h}', \mathbf{h}')] dx \\ &= \int_{x=a}^b \left[\left(\left[\nabla_{\mathbf{y}, \mathbf{y}}^2 F(x, \mathbf{y}, \mathbf{y}') - \frac{d}{dx} \nabla_{\mathbf{y}, \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \right] \mathbf{h}, \mathbf{h} \right) \right. \\ &\quad \left. + (\nabla_{\mathbf{y}', \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \mathbf{h}', \mathbf{h}') \right] dx \end{aligned}$$

if we apply the vector version of Lemma 22.7. $\square$

The next theorem gives the *sufficient conditions* for the vector function $\mathbf{y}$ to be a minimizer in the problem (22.22).

Theorem 22.8. (The Jacobi vector form) *If for some vector function $\mathbf{y} \in (C^1[a, b])^n$, verifying (22.23), the following conditions are fulfilled:*

1. *it satisfies the **Euler–Lagrange necessary condition** (22.24);*
2. *it satisfies the **strong Legendre vector condition***

$$\boxed{\nabla_{\mathbf{y}', \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \geq k I_{n \times n}, \quad k > 0} \quad (22.26)$$

3. *there exists vector functions $\mathbf{u}^i \in (C^1[a, b])^n$ ($i = 1, \dots, n$), such that*

$$\boxed{\begin{aligned} &\det [\mathbf{u}^1(x) \cdots \mathbf{u}^n(x)] \neq 0 \quad \text{for all } x \in [a, b] \\ &\mathbf{u}^i(a) = 0, \quad \frac{d}{dx} \mathbf{u}^i(a) = \mathbf{e}^i := \left(\underbrace{0, \dots, 1, 0, \dots, 0}_i \right)^\tau \end{aligned}} \quad (22.27)$$

*and satisfying the next ODE (the **Jacobi vector form ODE**):*

$$\boxed{\mathbf{Q}\mathbf{u} - \frac{d}{dx}(\mathbf{P}\mathbf{u}') = 0} \quad (22.28)$$

with

$$\begin{aligned} \mathbf{Q} &:= \nabla_{\mathbf{y}, \mathbf{y}}^2 F(x, \mathbf{y}, \mathbf{y}') - \frac{d}{dx} \nabla_{\mathbf{y}, \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \\ \mathbf{P} &:= \nabla_{\mathbf{y}', \mathbf{y}'}^2 F(x, \mathbf{y}, \mathbf{y}') \end{aligned}$$

then this vector function provides the local minimum to the functional $J(\mathbf{y})$.

Proof. It follows directly from the vector form of Theorem 22.2 if we define $\|\mathbf{h}\|_{\mathcal{B}}^2$ for $\mathcal{B} = (C^1[a, b])^n$ (see (18.6)) as

$$\|\mathbf{h}\|_{(C^1[a, b])^n}^2 := \sum_{i=1}^n \left(\max_{x \in [a, b]} |h^i(x)| + \max_{x \in [a, b]} \left| \frac{d}{dx} h^i(x) \right| \right)$$

and, practically, repeat the proof of Theorem 22.5. $\square$

22.4.1.3 Integral functional depending on derivatives of an order more than one

Let us consider the first-type necessary condition of the optimality for the integral functions

$$J(y) = \int_{x=a}^b F(x, y, y^{(1)}, \dots, y^{(n)}) dx \quad (22.29)$$

within the functions $y \in C^n[a, b]$, satisfying the boundary conditions

$$y^{(i)}(a) = \alpha_i, \quad y^{(i)}(b) = \beta_i \quad (i = 1, \dots, n) \quad (22.30)$$

First, notice that

$$\varphi_1(y, h) = \int_{x=a}^b \left[\sum_{i=0}^n \frac{\partial F}{\partial y^{(i)}} h^{(i)} \right] dx, \quad y^{(0)} := y$$

where the variation functions $h^{(i)}$ satisfy the conditions

$$h^{(i)}(a) = h^{(i)}(b) = 0 \quad (i = 1, \dots, n)$$

Integrating each integral term $\int_{x=a}^b \frac{\partial F}{\partial y^{(i)}} h^{(i)} dx$ of this equality i -times we derive

$$\varphi_1(y, h) = \int_{x=a}^b \left[\frac{\partial F}{\partial y} + \sum_{i=1}^n (-1)^i \frac{d}{dx^i} \frac{\partial F}{\partial y^{(i)}} \right] h dx \quad (22.31)$$

Theorem 22.9. (Euler–Poisson) *Let the curve $y \in C^n[a, b]$, satisfying (22.30), be an extremal (minimum or maximum) curve for the functional (22.29). Then it should satisfy the following ODE:*

$$\frac{\partial F}{\partial y} + \sum_{i=1}^n (-1)^i \frac{d}{dx^i} \frac{\partial F}{\partial y^{(i)}} = 0 \quad (22.32)$$

Proof. It follows directly from (22.31) applying the Lagrange lemma 22.5. $\square$

22.4.2 Curves with non-fixed boundary points

Problem formulation: among all smooth curves $y = y(x)$ with the boundary points (x_0, y_0) and (x_1, y_1) , belonging to the given curves ("sliding surfaces") $y = \varphi(x)$ and $y = \psi(x)$, find one which minimizes the functional

$$J(y) = \int_{x=x_0: y(x_0)=\varphi(x_0)}^{x_1: y(x_1)=\psi(x_1)} F(x, y(x), y'(x)) dx \quad (22.33)$$

Realizing the function variation $(y + h)$ such that

$$\begin{aligned} h(x_i) &\simeq \delta y(x_i) - y'(x_i) \delta x_i, \quad i = 0, 1 \\ \delta y(x_0) &\simeq \varphi'(x_0) \delta x_0, \quad \delta y(x_1) \simeq \psi'(x_1) \delta x_1 \end{aligned} \quad (22.34)$$

and integrating by parts, we derive

$$\begin{aligned} \varphi_1(y_0, h) &= \int_{x=x_0}^{x_1} \left(\frac{\partial F}{\partial y} h + \frac{\partial F}{\partial y'} h' \right) dx + F(x, y, y') \delta x \Big|_{x=x_0}^{x=x_1} \\ &= \int_{x=x_0}^{x_1} \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) h dx + \frac{\partial F}{\partial y'} h \Big|_{x=x_0}^{x=x_1} + F(x, y, y') \delta x \Big|_{x=x_0}^{x=x_1} \end{aligned} \quad (22.35)$$

Remark 22.1. Since the problem in 22.4.2 contains as a partial case the problem (22.9) with fixed boundary points, then any solution of the problem in 22.4.2 should satisfy the Euler–Lagrange condition (22.13), which simplifies (22.35) up to

$$\varphi_1(y_0, h) = \frac{\partial F}{\partial y'} h \Big|_{x=x_0}^{x=x_1} + F(x, y, y') \delta x \Big|_{x=x_0}^{x=x_1} \quad (22.36)$$

Applications (22.34) to (22.36) imply

$$\begin{aligned} \varphi_1(y_0, h) &= \frac{\partial F}{\partial y'} \delta y \Big|_{x=x_0}^{x=x_1} + \left(F - \frac{\partial F}{\partial y'} y' \right) \delta x \Big|_{x=x_0}^{x=x_1} \\ &= \left(\frac{\partial F}{\partial y'} \psi' + F - \frac{\partial F}{\partial y'} y' \right) \Big|_{x=x_1} \delta x_1 \\ &\quad + \left(\frac{\partial F}{\partial y'} \varphi' + F - \frac{\partial F}{\partial y'} y' \right) \Big|_{x=x_0} \delta x_0 \end{aligned} \quad (22.37)$$

Theorem 22.10. If some curve $y = y(x) \in C^1$ with the boundary points (x_0, y_0) and (x_1, y_1) , belonging to the given curves ("sliding surfaces") $y = \varphi(x)$ and $y = \psi(x)$ provides an extremum to the functional (22.33), then

1. it satisfies the **Euler–Lagrange condition** (22.13), i.e.,

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0$$

2. it should, additionally, satisfy the, so-called, **transversality conditions**

$$\boxed{\begin{aligned} \left[F + \frac{\partial F}{\partial y'} (\psi' - y') \right] \Big|_{x=x_1} &= 0 \\ \left[F + \frac{\partial F}{\partial y'} (\psi' - y') \right] \Big|_{x=x_0} &= 0 \end{aligned}} \quad (22.38)$$

Proof. The relations (22.38) result from (22.37) if we take into account that the variations δx_0 and δx_1 are independent. $\square$

Example 22.3.

$$J(y) = \int_{x_0=0: y(x_0)=0}^{x_1: y(x_1)=2-x} \frac{\sqrt{1 + (y'(x))^2}}{x} dx$$

The Euler–Lagrange condition (22.13) is $-\frac{d}{dx} \frac{y'}{x \sqrt{1 + (y')^2}} = 0$, which leads to the following relation

$$\begin{aligned} \frac{y'}{x \sqrt{1 + (y')^2}} &= c_1, \quad y' = \pm \frac{xc_1}{\sqrt{1 - x^2 c_1^2}} \\ y &= \mp \frac{1}{c_1} \sqrt{1 - x^2 c_1^2} + c_2, \quad (y - c_2)^2 = 1/c_1^2 - x^2 \end{aligned}$$

The boundary condition $y(0) = 0$ gives $c_1^2 = 1/c_2^2$. So, finally, the Euler–Lagrange condition (22.13) is $(y - c_2)^2 + x^2 = c_2^2$. The transversality conditions (22.38) imply $0 = y' = -\frac{x}{y - c_2} = 1$, which, together with $y = 2 - x$, gives $c_2 = 2$. Finally, the extremal curve is as follows:

$$(y - 2)^2 + x^2 = 4$$

22.4.3 Curves with a nonsmoothness point

If an extremal curve has a nonsmooth point $x^* \in [a, b]$ in the problem (22.9) with a fixed boundary point, that is,

$$y(x) \text{ is continuous in } x^*$$

$$y'(x) \Big|_{x \rightarrow x^*-0} \neq y'(x) \Big|_{x \rightarrow x^*+0}$$

then

$$\varphi_1(y_0, h) = \varphi_{1,1}(y_0, h) + \varphi_{1,2}(y_0, h) = 0$$

where (we can consider at each semi-interval $[a, x^*]$ and $[x^*, b]$ the boundary point x^* as a non-fixed one)

$$\begin{aligned}\varphi_{1,1}(y_0, h) &= \frac{\partial F}{\partial y'} \delta y|_{x=x^*-0} + \left(F - \frac{\partial F}{\partial y'} y' \right) \delta x|_{x=x^*-0} \\ \varphi_{1,2}(y_0, h) &= -\frac{\partial F}{\partial y'} \delta y|_{x=x^*+0} - \left(F - \frac{\partial F}{\partial y'} y' \right) \delta x|_{x=x^*+0}\end{aligned}$$

which gives

$$\begin{aligned}0 &= \varphi_1(y_0, h) \\ &= \left(\frac{\partial F}{\partial y'}|_{x=x^*-0} - \frac{\partial F}{\partial y'}|_{x=x^*+0} \right) \delta y \\ &\quad + \left[\left(F - \frac{\partial F}{\partial y'} y' \right)|_{x=x^*-0} - \left(F - \frac{\partial F}{\partial y'} y' \right)|_{x=x^*+0} \right] \delta x\end{aligned}$$

Since here δy and δx are admitted to be arbitrary, we obtain the, so-called, *Weierstrass–Erdmann conditions*:

$$\boxed{\begin{aligned}\frac{\partial F}{\partial y'}|_{x=x^*-0} &= \frac{\partial F}{\partial y'}|_{x=x^*+0} \\ \left(F - \frac{\partial F}{\partial y'} y' \right)|_{x=x^*-0} &= \left(F - \frac{\partial F}{\partial y'} y' \right)|_{x=x^*+0}\end{aligned}} \quad (22.39)$$

Example 22.4. Consider the functional

$$J(y) = \int_{x=a}^b (y')^2 (1 - y')^2 dx$$

The boundary points are assumed to be fixed. The Euler–Lagrange condition (22.13) gives $y = c_1 x + c_2$. The *Weierstrass–Erdmann conditions* (22.39) are

$$\begin{aligned}2y'(1 - y')(1 - 2y')|_{x=x^*-0} &= 2y'(1 - y')(1 - 2y')|_{x=x^*+0} \\ -(y')^2(1 - y')(1 - 3y')|_{x=x^*-0} &= -(y')^2(1 - y')(1 - 3y')|_{x=x^*+0}\end{aligned}$$

which are fulfilled for extremal curves such that

$$y'|_{x=x^*-0} = 0, \quad y'|_{x=x^*+0} = 1 \quad \text{or} \quad y'|_{x=x^*-0} = 1, \quad y'|_{x=x^*+0} = 0$$

22.5 Optimal control problem

22.5.1 Controlled plant, cost functionals and terminal set

Consider the controlled plant given by the following system of ordinary differential equations (ODE)

$$\left. \begin{array}{l} x(t) = f(x(t), u(t), t), \quad \text{a.e. } t \in [0, T] \\ x(0) = x_0 \end{array} \right\} \quad (22.40)$$

where $x = (x^1, \dots, x^n)^T \in \mathbb{R}^n$ is its state vector, $u = (u^1, \dots, u^r)^T \in \mathbb{R}^r$ is the control that may run over a given control region $U \subset \mathbb{R}^r$ with the cost functional

$$J(u(\cdot)) := h_0(x(T)) + \int_{t=0}^T h(x(t), u(t), t) dt \quad (22.41)$$

containing the integral term as well as the terminal one and with the terminal set $\mathcal{M} \subseteq \mathbb{R}^n$ given by the inequalities

$$\mathcal{M} = \{x \in \mathbb{R}^n : g_l(x) \leq 0 \quad (l = 1, \dots, L)\} \quad (22.42)$$

The time process or *horizon* T is supposed to be fixed or nonfixed and may be finite or infinite.

Definition 22.4.

- (a) The function (22.41) is said to be given in **Bolza form**.
 (b) If in (22.41) $h_0(x) = 0$ we obtain the cost functional in **Lagrange form**, that is,

$$J(u(\cdot)) = \int_{t=0}^T h(x(t), u(t), t) dt \quad (22.43)$$

- (c) If in (22.41) $h(x, u, t) = 0$ we obtain the cost functional in **Mayer form**, that is,

$$J(u(\cdot)) = h_0(x(T)) \quad (22.44)$$

Usually the following assumptions are assumed to be in force:

- (A1) (U, d) is a separable metric space (with the metric d) and $T > 0$.
 (A2) The maps

$$\left. \begin{array}{l} f : \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R}^n \\ h : \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R} \\ h_0 : \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R} \\ g_l : \mathbb{R}^n \rightarrow \mathbb{R}^n \quad (l = 1, \dots, L) \end{array} \right\} \quad (22.45)$$

are measurable and there exist a constant L and a continuity modulus $\bar{\omega} : [0, \infty) \rightarrow [0, \infty)$ such that for $\varphi = f(x, u, t)$, $h(x, u, t)$, $h_0(x, u, t)$, $g_l(x)$ ($l = 1, \dots, L$) the following inequalities hold:

$$\left. \begin{aligned} \|\varphi(x, u, t) - \varphi(\hat{x}, \hat{u}, t)\| &\leq L \|x - \hat{x}\| + \bar{\omega}(d(u, \hat{u})) \\ \forall t \in [0, T], \quad x, \hat{x} &\in \mathbb{R}^n, \quad u, \hat{u} \in U \\ \|\varphi(0, u, t)\| &\leq L \quad \forall u, t \in U \times [0, T] \end{aligned} \right\} \quad (22.46)$$

(A3) The maps f, h, h_0 and g_l ($l = 1, \dots, L$) are from C^1 in x and there exists a continuity modulus $\bar{\omega} : [0, \infty) \rightarrow [0, \infty)$ such that for $\varphi = f(x, u, t)$, $h(x, u, t)$, $h_0(x, u, t)$, $g_l(x)$ ($l = 1, \dots, L$) the following inequalities hold:

$$\left. \begin{aligned} \left\| \frac{\partial}{\partial x} \varphi(x, u, t) - \frac{\partial}{\partial x} \varphi(\hat{x}, \hat{u}, t) \right\| \\ \leq \bar{\omega}(\|x - \hat{x}\| + d(u, \hat{u})) \\ \forall t \in [0, T], \quad x, \hat{x} \in \mathbb{R}^n, \quad u, \hat{u} \in U \end{aligned} \right\} \quad (22.47)$$

22.5.2 Feasible and admissible control

Definition 22.5. A function $u(t)$, $t_0 \leq t \leq T$, is said to be

(a) a **feasible control** if it is measurable and $u(t) \in U$ for all $t \in [0, T]$. Denote the set of all feasible controls by

$$\mathcal{U}[0, T] := \{u(\cdot) : [0, T] \rightarrow U \mid u(t) \text{ is measurable}\} \quad (22.48)$$

(b) an **admissible or realizing the terminal condition** (22.42), if the corresponding trajectory $x(t)$ satisfies the terminal condition, that is, satisfies the inclusion $x(T) \in \mathcal{M}$. Denote the set of all admissible controls by

$$\mathcal{U}_{admis}[0, T] := \{u(\cdot) : u(\cdot) \in \mathcal{U}[0, T], \quad x(T) \in \mathcal{M}\} \quad (22.49)$$

In view of Theorem 19.1 on the existence and the uniqueness of an ODE solution, it follows that under assumptions (A1)–(A2) for any $u(t) \in \mathcal{U}[0, T]$ equation (1.4) admits a unique solution $x(\cdot) := x(\cdot, u(\cdot))$ and the functional (22.41) is well defined.

22.5.3 Problem setting in the general Bolza form

Based on the definitions given above, the optimal control problem (OCP) can be formulated as follows.

Problem 22.1. (OCP in Bolza form)

$$\text{Minimize (22.41) over } \mathcal{U}_{admis}[0, T] \quad (22.50)$$

Problem 22.2. (OCP with a fixed terminal term) *If in the problem (22.50)*

$$\mathcal{M} = \{x \in \mathbb{R}^n : g_1(x) = x - x_f \leq 0, \quad g_2(x) = -(x - x_f) \leq 0\} \quad (22.51)$$

*then it is called the **optimal control problem with fixed terminal term** x_f .*

Definition 22.6. *Any control $u^*(\cdot) \in \mathcal{U}_{admis}[0, T]$ satisfying*

$$J(u^*(\cdot)) = \inf_{u(\cdot) \in \mathcal{U}_{admis}[0, T]} J(u(\cdot)) \quad (22.52)$$

*is called an **optimal control**. The corresponding state trajectories $x^*(\cdot) := x^*(\cdot, u^*(\cdot))$ are called an **optimal state trajectory**, and $(x^*(\cdot), u^*(\cdot))$ is called an **optimal pair**.*

22.5.4 Mayer form representation

Summary 22.1. *Introduce $(n+1)$ -dimensional space $\mathbb{R}^{n+1}$ of the variables $x = (x_1, \dots, x_n, x_{n+1})^\top$ where the first n coordinates satisfy (22.40) and the component x_{n+1} is given by*

$$x_{n+1}(t) := \int_{\tau=0}^t h(x(\tau), u(\tau), \tau) d\tau \quad (22.53)$$

or, in the differential form,

$$\dot{x}_{n+1}(t) = h(x(t), u(t), t) \quad (22.54)$$

with the initial condition for the last component given by

$$x_{n+1}(0) = 0 \quad (22.55)$$

As a result, the initial optimization problem in the Bolza form (22.50) can be reformulated in the space $\mathbb{R}^{n+1}$ as the Mayer problem with the cost functional $J(u(\cdot))$

$$J(u(\cdot)) = h_0(x(T)) + x_{n+1}(T) \quad (22.56)$$

where the function $h_0(x)$ does not depend on the last coordinate $x_{n+1}(t)$, that is,

$$\frac{\partial}{\partial x_{n+1}} h_0(x) = 0 \quad (22.57)$$

Summary 22.2. *From the relations above it follows that the Mayer problem with the cost function (22.56) is equivalent to the initial optimization control problem (22.50) in the Bolza form.*

There exist two principal approaches to solving *optimal control problems*:

- the first one is the *maximum principle* (MP) of L. Pontryagin (Boltyanski *et al.* 1956; Pontryagin *et al.* 1969 (translated from Russian));

- and the second one is the *dynamic programming method* (DPM) of R. Bellman (1957).

We will touch on both of them below.

22.6 Maximum principle

The *maximum principle* is a basic instrument to derive a set of *necessary conditions* which should satisfy any optimal control. As an optimal control problem may be regarded as an optimization problem in the corresponding infinite dimensional (Hilbert or, in general, Banach) space, the necessary conditions (resembling the *Kuhn–Tucker conditions* in the finite-dimensional optimization) take place. They are known as the *maximum principle* which is really a milestone in modern optimal control theory. It states that any dynamic system closed by an optimal control strategy or, simply, by an optimal control is a Hamiltonian (with the doubled dimension) system given by a system of the forward–backward ordinary differential equations and, in addition, an optimal control maximizes the function called Hamiltonian. Its mathematical importance consists of the following fact: the maximization of the Hamiltonian with respect to a control variable given in a finite-dimensional space looks and really is much easier than the original optimization problem formulated in an infinite-dimensional space. The key idea of the original version of the maximum principle comes from classical variations calculus. To derive the main MP formulation, first one needs to perturb slightly an optimal control using the so-called needle-shape (spike) variations and, second, to consider the first-order term in a Taylor expansion with respect to this perturbation. Tending perturbations to zero, some variation inequalities may be obtained. Then the final result follows directly from duality.

22.6.1 Needle-shape variations

Let $(x^*(\cdot), u^*(\cdot))$ be the given optimal pair for the problem (22.52) and $M_\varepsilon \subseteq [0, T]$ be a measurable set of the time interval with Lebesgue measure $|M_\varepsilon| = \varepsilon > 0$. Let now $u(\cdot) \in \mathcal{U}_{admis} [0, T]$ be any given admissible control.

Definition 22.7. Define the following control

$$u^\varepsilon(t) := \begin{cases} u^*(t) & \text{if } t \in [0, T] \setminus M_\varepsilon \\ u(t) \in \mathcal{U}_{admis} [0, T] & \text{if } t \in M_\varepsilon \end{cases} \quad (22.58)$$

Evidently $u^\varepsilon(\cdot) \in \mathcal{U}_{admis} [0, T]$. Below $u^\varepsilon(\cdot)$ is referred to as a **needle-shape** or **spike variation** of the optimal control $u^*(t)$.

The next lemma plays a key role in proving the basic MP theorem.

Lemma 22.8. (The variational equation) Let $x^\varepsilon(\cdot) := x(\cdot, u^\varepsilon(\cdot))$ be the solution of (22.52) for the plant model given by (22.40) under the control $u^\varepsilon(\cdot)$ and $\Delta^\varepsilon(\cdot)$ be the

solution to the following differential equation

$$\begin{aligned}\dot{\Delta}^\varepsilon(t) &= \frac{\partial}{\partial x} f(x^*(t), u^*(t), t) \Delta^\varepsilon(t) \\ &\quad + [f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t)\end{aligned}\quad (22.59)$$

$$\Delta^\varepsilon(0) = 0$$

where $\chi_{M_\varepsilon}(t)$ is the characteristic function of the set M_ε , that is

$$\chi_{M_\varepsilon}(t) := \begin{cases} 1 & \text{if } t \in M_\varepsilon \\ 0 & \text{if } t \notin M_\varepsilon \end{cases} \quad (22.60)$$

Then

$$\left. \begin{aligned} \max_{t \in [0, T]} \|x^\varepsilon(t) - x^*(t)\| &= O(\varepsilon) \\ \max_{t \in [0, T]} \|\Delta^\varepsilon(t)\| &= O(\varepsilon) \\ \max_{t \in [0, T]} \|x^\varepsilon(t) - x^*(t) - \Delta^\varepsilon(t)\| &= o(\varepsilon) \end{aligned} \right\} \quad (22.61)$$

and the following variational equations hold

(a) for the cost function given in Bolza form (22.41)

$$\begin{aligned} J(u^\varepsilon(\cdot)) - J(u^*(\cdot)) &= \left(\frac{\partial}{\partial x} h_0(x^*(T)), \Delta^\varepsilon(T) \right) \\ &\quad + \int_0^T \left\{ \left(\frac{\partial}{\partial x} h(x^*(t), u^*(t), t), \Delta^\varepsilon(t) \right) \right. \\ &\quad \left. + [h(x^*(t), u^\varepsilon(t), t) - h(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t) \right\} dt + o(\varepsilon) \end{aligned} \quad (22.62)$$

(b) for the cost function given in Mayer form (22.44)

$$J(u^\varepsilon(\cdot)) - J(u^*(\cdot)) = \left(\frac{\partial}{\partial x} h_0(x^*(T)), \Delta^\varepsilon(T) \right) + o(\varepsilon) \quad (22.63)$$

Proof. Define $\delta^\varepsilon(t) := x^\varepsilon(t) - x^*(t)$. Then assumption (A2) (22.46) for any $t \in [0, T]$ implies

$$\|\delta^\varepsilon(t)\| \leq \int_0^t L \|\delta^\varepsilon(s)\| ds + K\varepsilon \quad (22.64)$$

which, by the Gronwall lemma 19.4, leads to the first relation in (22.61). Define

$$\eta^\varepsilon(t) := x^\varepsilon(t) - x^*(t) - \Delta^\varepsilon(t) = \delta^\varepsilon(t) - \Delta^\varepsilon(t) \quad (22.65)$$

Then

$$\begin{aligned}
 \dot{\eta}^\varepsilon(t) &= [f(x^\varepsilon(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \\
 &\quad - \frac{\partial}{\partial x} f(x^*(t), u^*(t), t) \Delta^\varepsilon(t) \\
 &\quad - [f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t) \\
 &= \int_{\theta=0}^1 \left[\frac{\partial}{\partial x} f(x^*(t) + \theta \delta^\varepsilon(t), u^\varepsilon(t), t) - \frac{\partial}{\partial x} f(x^*(t), u^*(t), t) \right] d\theta \delta^\varepsilon(t) \\
 &\quad - [f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t) \\
 &\quad + \frac{\partial}{\partial x} f(x^*(t), u^*(t), t) \eta^\varepsilon(t)
 \end{aligned} \tag{22.66}$$

Integrating the last identity (22.66) and in view of (A2) (22.46) and (A3) (22.47), we obtain

$$\begin{aligned}
 \|\eta^\varepsilon(t)\| &\leq \int_{s=0}^t \int_{\theta=0}^1 [\bar{\omega}(\theta \|\delta^\varepsilon(s)\| + d(u^\varepsilon(s), u^*(s))) \|\delta^\varepsilon(s)\| d\theta ds \\
 &\quad + \int_{s=0}^t \bar{\omega}(d(u^\varepsilon(s), u^*(s))) \chi_{M_\varepsilon}(s) ds + \int_{s=0}^t \frac{\partial}{\partial x} f(x^*(s), u^*(s), s) \eta^\varepsilon(s) ds \\
 &\leq \text{Const} \cdot \varepsilon \cdot o(1) + \text{Const} \cdot \int_{s=0}^t \|\eta^\varepsilon(s)\| ds
 \end{aligned} \tag{22.67}$$

The last inequality in (22.67) by the Gronwall lemma directly implies the third relation in (22.61). The second relation follows from the first and third ones. The same manipulations lead to (22.62) and (22.63). $\square$

22.6.2 Adjoint variables and MP formulation

The classical format of MP formulation gives a set of *first-order necessary conditions* for optimal pairs.

Theorem 22.11. (MP for Mayer form with a fixed horizon) *If under assumptions (A1)–(A3) a pair $(x^*(\cdot), u^*(\cdot))$ is optimal then there exist the vector functions $\psi(t)$, satisfying the system of the **adjoint equations***

$$\dot{\psi}(t) = - \frac{\partial}{\partial x} f(x^*(t), u^*(t), t)^\top \psi(t) \quad a.e. \quad t \in [0, T] \tag{22.68}$$

and nonnegative constants $\mu \geq 0$ and $v_l \geq 0$ ($l = 1, \dots, L$) such that the following four conditions hold:

(a) (**the maximality condition**): for almost all $t \in [0, T]$

$$H(\psi(t), x^*(t), u^*(t), t) = \max_{u \in U} H(\psi(t), x^*(t), u, t) \quad (22.69)$$

where the Hamiltonian is defined by

$$H(\psi, x, u, t) := \psi^\top f(x, u, t) \quad (22.70)$$

$$t, x, u, \psi \in [0, T] \times \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R}^n$$

(b) (**transversality condition**): the equality

$$\psi(T) + \mu \frac{\partial}{\partial x} h_0(x^*(T)) + \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x^*(T)) = 0 \quad (22.71)$$

holds;

(c) (**complementary slackness conditions**): either the equality $g_l(x^*(T)) = 0$ holds, or $v_l = 0$, that is, for any ($l = 1, \dots, L$)

$$v_l g_l(x^*(T)) = 0 \quad (22.72)$$

(d) (**nontriviality condition**): at least one of the numbers $|\psi(T)|$ and v_l is distinct from zero, that is,

$$\|\psi(T)\| + \mu + \sum_{l=1}^L v_l > 0 \quad (22.73)$$

Proof. Let $\psi(t)$ be the solution of (22.68) corresponding to the terminal condition $\psi(T) = b$ and $\bar{t} \in [0, T]$. Define $M_\varepsilon := [\bar{t}, \bar{t} + \varepsilon] \subseteq [0, T]$. If $u^*(t)$ is an optimal control, then according to the Lagrange principle (see Theorem 21.12), formulated for a Banach space, there exist constants $\mu \geq 0$ and $v_l \geq 0$ ($l = 1, \dots, L$) such that for any $\varepsilon \geq 0$

$$\mathcal{L}(u^\varepsilon(\cdot), \mu, v) - \mathcal{L}(u^*(\cdot), \mu, v) \geq 0 \quad (22.74)$$

where

$$\mathcal{L}(u(\cdot), \mu, v) := \mu J(u(\cdot)) + \sum_{l=1}^L v_l g_l(x(T)) \quad (22.75)$$

Taking into account that $\psi(T) = b$ and $\Delta^\varepsilon(0) = 0$, by the differential chain rule, applied to the term $\psi(t)^\top \Delta^\varepsilon(t)$, and, in view of (22.59) and (22.68), we obtain

$$b^\top \Delta^\varepsilon(T) = \psi(T)^\top \Delta^\varepsilon(T) - \psi(0)^\top \Delta^\varepsilon(0)$$

$$= \int_{t=0}^T d(\psi(t)^\top \Delta^\varepsilon(t)) = \int_{t=0}^T (\dot{\psi}(t)^\top \Delta^\varepsilon(t) + \psi(t)^\top \dot{\Delta}^\varepsilon(t)) dt$$

$$\begin{aligned}
&= \int_{t=0}^T \left[-\Delta^\varepsilon(t)^\top \frac{\partial}{\partial x} f(x^*(t), u^*(t), t)^\top \psi(t) \right. \\
&\quad + \psi(t)^\top \frac{\partial}{\partial x} f(x^*(t), u^*(t), t) \Delta^\varepsilon(t) \\
&\quad \left. + \psi(t)^\top [f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t) \right] dt \\
&= \int_{t=0}^T \psi(t)^\top [f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)] \chi_{M_\varepsilon}(t) dt
\end{aligned} \tag{22.76}$$

The variational equality (22.62) together with (22.74) and (22.76) implies

$$\begin{aligned}
0 &\leq \mathcal{L}(u^\varepsilon(\cdot), \mu, \nu) - \mathcal{L}(u^*(\cdot), \mu, \nu) \\
&= \mu \left(\frac{\partial}{\partial x} h_0(x^*(T)), \Delta^\varepsilon(T) \right) + b^\top \Delta^\varepsilon(T) \\
&\quad - \int_{t=0}^T \psi(t)^\top (f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t)) \chi_{M_\varepsilon}(t) dt \\
&\quad + \sum_{l=1}^L \nu_l [g_l(x(T)) - g_l(x^*(T))] + o(\varepsilon) \\
&= \left(\mu \frac{\partial}{\partial x} h_0(x^*(T)) + b + \sum_{l=1}^L \nu_l \frac{\partial}{\partial x} g_l(x^*(T)), \Delta^\varepsilon(T) \right) \\
&\quad - \int_{t=\bar{t}}^{\bar{t}+\varepsilon} [\psi(t)^\top (f(x^*(t), u^\varepsilon(t), t) - f(x^*(t), u^*(t), t))] dt + o(\varepsilon) \\
&= \left(\mu \frac{\partial}{\partial x} h_0(x^*(T)) + b + \sum_{l=1}^L \nu_l \frac{\partial}{\partial x} g_l(x^*(T)), \Delta^\varepsilon(T) \right) \\
&\quad - \int_{t=\bar{t}}^{\bar{t}+\varepsilon} [H(\psi(t), x^*(t), u^\varepsilon(t), t) - H(\psi(t), x^*(t), u^*(t), t)] dt
\end{aligned} \tag{22.77}$$

1. Tending ε to zero from (22.77) it follows that

$$0 \leq \left(\mu \frac{\partial}{\partial x} h_0(x^*(T)) + b + \sum_{l=1}^L \nu_l \frac{\partial}{\partial x} g_l(x^*(T)), \Delta^\varepsilon(T) \right) \Big|_{\varepsilon=0}$$

which should be valid for any $\Delta^\varepsilon(T)|_{\varepsilon=0}$. This is possible only if (this can be proved by contradiction)

$$\mu \frac{\partial}{\partial x} h_0(x^*(T)) + b + \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x^*(T)) = 0 \quad (22.78)$$

which is equivalent to (22.71). So, the transversality condition is proven.

2. In view of (22.78), the inequality (22.77) may be simplified to the following one

$$0 \leq - \int_{t=\bar{t}}^{\bar{t}+\varepsilon} [H(\psi(t), x^*(t), u^\varepsilon(t), t) - H(\psi(t), x^*(t), u^*(t), t)] dt \quad (22.79)$$

This inequality together with separability of metric space U directly leads to the maximality condition (22.69).

3. Suppose that (22.72) does not hold, that is, there exist an index l_0 and a multiplier $\tilde{v}_{l_0}$ such that $v_{l_0} g_{l_0}(x^*(T)) < 0$. This gives

$$\begin{aligned} \mathcal{L}(u^*(\cdot), \mu, \tilde{v}) &:= \mu J(u^*(\cdot)) + \sum_{l=1}^L \tilde{v}_l g_l(x^*(T)) \\ &= \mu J(u^*(\cdot)) + \tilde{v}_{l_0} g_{l_0}(x^*(T)) < \mu J(u^*(\cdot)) = \mathcal{L}(u^*(\cdot), \mu, v) \end{aligned}$$

It means that $u^*(\cdot)$ is not optimal control. We obtain the contradiction. So, the complementary slackness condition is proven too.

4. Suppose that (22.73) is not valid, i.e., $|\psi(T)| + \mu + \sum_{l=1}^L v_l = 0$. This implies $\psi(T) = 0, \mu = v_l = 0 (l = 1, \dots, L)$, and, hence, in view of (22.68) and by the Gronwall lemma 19.4, it follows that $\psi(t) = 0$ for all $t \in [0, T]$. So, $H(\psi(t), x(t), u(t), t) = 0$ for any $u(t)$ (not only for $u^*(t)$). This means that the application of any admissible control keeps the cost function unchangeable which corresponds to a trivial situation. So, the nontriviality condition is proven too. $\square$

22.6.3 The regular case

In the, so-called, *regular case*, when $\mu > 0$ (this means that the nontriviality condition holds automatically), the variable $\psi(t)$ and constants v_l may be normalized and change to $\tilde{\psi}(t) := \psi(t)/\mu$ and $\tilde{v}_l := v_l/\mu$. In this new variable the MP formulation looks as follows.

Theorem 22.12. (MP in the regular case) *If under assumptions (A1)–(A3) a pair $(x^*(\cdot), u^*(\cdot))$ is optimal, then there exists a vector function $\tilde{\psi}(t)$ satisfying the system of the adjoint equations*

$$\frac{d}{dt} \tilde{\psi}(t) = - \frac{\partial}{\partial x} f(x^*(t), u^*(t), t)^\top \tilde{\psi}(t) \quad \text{a.e. } t \in [0, T]$$

and $v_l \geq 0$ ($l = 1, \dots, L$) such that the following three conditions hold:

(a) (**the maximality condition**): for almost all $t \in [0, T]$

$$H(\tilde{\psi}(t), x^*(t), u^*(t), t) = \max_{u \in U} H(\tilde{\psi}(t), x^*(t), u, t) \quad (22.80)$$

where the Hamiltonian is defined by

$$H(\psi, x, u, t) := \tilde{\psi}^\top f(x, u, t)$$

$$t, x, u, \psi \in [0, T] \times \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R}^n$$

(b) (**transversality condition**): for every $\alpha \in \mathcal{A}$, the equalities

$$\tilde{\psi}(T) + \frac{\partial}{\partial x} h_0(x^*(T)) + \sum_{l=1}^L \tilde{v}_l \frac{\partial}{\partial x} g_l(x^*(T)) = 0$$

hold;

(c) (**complementary slackness conditions**): either the equality $g_l(x^*(T)) = 0$ holds, or $v_l = 0$, that is, for any ($l = 1, \dots, L$) $v_l g_l(x^*(T)) = 0$.

Remark 22.2. This means that without loss of generality we may put $\mu = 1$. It may be shown that the regularity property takes place if the vectors $\frac{\partial}{\partial x} g_l(x^*(T))$ are linearly independent. The verification of this property is usually not so simple a task.

22.6.4 Hamiltonian form and constancy property

Corollary 22.3. (Hamiltonian for the Bolza problem) Hamiltonian for the Bolza problem has the form

$$\begin{aligned} H(\psi, x, u, t) &:= \psi^\top f(x, u, t) - \mu h(x(t), u(t), t) \\ t, x, u, \psi &\in [0, T] \times \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R}^n \end{aligned} \quad (22.81)$$

Proof. It follows from (22.53)–(22.57). Indeed, since the Mayer's form representation ($\dot{\psi}_{n+1}(t) = h(x(t), u(t), t)$) implies $\dot{\psi}_{n+1}(t) = 0$, then $\psi_{n+1}(T) = -\mu$. $\square$

Corollary 22.4. (Hamiltonian form) Equations (22.40) and (22.68) may be represented in the, so-called, Hamiltonian form (the forward-backward ODE form):

$$\left. \begin{aligned} \dot{x}^*(t) &= \frac{\partial}{\partial \psi} H(\psi(t), x^*(t), u^*(t), t), \quad x^*(0) = x_0 \\ \dot{\psi}(t) &= -\frac{\partial}{\partial x} H(\psi(t), x^*(t), u^*(t), t) \\ \psi(T) &= -\mu \frac{\partial}{\partial x} h_0(x^*(T)) - \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x^*(T)) \end{aligned} \right\} \quad (22.82)$$

Proof. It directly follows from comparison of the right-hand side of (22.70) with (22.40) and (22.68). $\square$

Corollary 22.5. (Constancy property) For stationary systems when in (22.40), (22.41)

$$f = f(x(t), u(t)), \quad h = h(x(t), u(t)) \quad (22.83)$$

it follows that for all $t \in [t_0, T]$

$$H(\psi(t), x^*(t), u^*(\psi(t), x^*(t))) = \text{const} \quad (22.84)$$

Proof. One can see that in this case the Hamiltonian $H = H(\psi(t), x(t), u(t))$ does not depend on t directly, that is, $\frac{\partial}{\partial t} H = 0$. Hence, $u^*(t)$ is a function of $\psi(t)$ and $x^*(t)$ only, i.e., $u^*(t) = u^*(\psi(t), x^*(t))$. Denote

$$H(\psi(t), x^*(t), u^*(\psi(t), x^*(t))) := \tilde{H}(\psi(t), x^*(t))$$

Then (22.82) becomes

$$\dot{x}(t) = \frac{\partial}{\partial \psi} \tilde{H}(\psi(t), x^*(t)), \quad \dot{\psi}(t) = -\frac{\partial}{\partial x} \tilde{H}(\psi(t), x^*(t))$$

which implies

$$\begin{aligned} \frac{d}{dt} \tilde{H}(\psi(t), x^*(t)) &= \frac{\partial}{\partial \psi} \tilde{H}(\psi(t), x^*(t))^T \dot{\psi}(t) \\ &\quad + \frac{\partial}{\partial x} \tilde{H}(\psi(t), x^*(t))^T \dot{x}(t) = 0 \end{aligned}$$

and hence $\tilde{H}(\psi(t), x^*(t)) = \text{const}$ for any $t \in [t_0, T]$. □

22.6.5 Nonfixed horizon optimal control problem and zero property

Consider the following generalization of the optimal control problem (22.40), (22.44), (22.50) permitting terminal time to be free. In view of this, the optimization problem may be formulated in the following manner: *minimize*

$$J(u(\cdot)) = h_0(x(T), T) \quad (22.85)$$

over $u(\cdot) \in \mathcal{U}_{\text{admis}}[0, T]$ and $T \geq 0$ with the terminal set $\mathcal{M}(T)$ given by

$$\mathcal{M}(T) = \{x(T) \in \mathbb{R}^n : g_l(x(T), T) \leq 0 \quad (l = 1, \dots, L)\} \quad (22.86)$$

Theorem 22.13. (MP for non fixed horizon case) If under assumptions (A1)–(A3) the pair $(T^*, u^*(\cdot))$ is a solution of the problem (22.85), (22.86) and $x^*(t)$ is the corresponding optimal trajectory, then there exist the vector functions $\psi(t)$, satisfying the system of the adjoint equations (22.68), and nonnegative constants $\mu \geq 0$ and $v_l \geq 0$

($l = 1, \dots, L$) such that all four conditions of the previous Theorem 22.11 are fulfilled and, in addition, the following condition to the terminal time holds:

$$\boxed{\begin{aligned} H(\psi(T), x(T), u(T), T) &:= \psi^\top(t) f(x(T), u(T-0), T) \\ &= \mu \frac{\partial}{\partial T} h_0(x^*(T), T) + \sum_{l=1}^L v_l \frac{\partial}{\partial T} g_l(x^*(T), T) \end{aligned}} \quad (22.87)$$

Proof. Since $(T^*, u^*(\cdot))$ is a solution of the problem then evidently $u^*(\cdot)$ is a solution of the problem (22.40), (22.44), (22.50) with the fixed horizon $T = T^*$ and, hence, all four properties of the Theorem 22.11 with $T = T^*$ should be fulfilled. Let us find the additional condition to the terminal time T^* which should be satisfied too.

(a) Consider again the needle-shape variation defined as

$$u^\varepsilon(t) := \begin{cases} u^*(t) & \text{if } t \in [0, T^*] \setminus (M_\varepsilon \cap (T^* - \varepsilon, T^*]) \\ u(t) \in \mathcal{U}_{\text{admis}}[0, T^*] & \text{if } t \in M_\varepsilon \subseteq [0, T^* - \varepsilon) \\ u(t) \in \mathcal{U}_{\text{admis}}[0, T^*] & \text{if } t \in [T^* - \varepsilon, T^*] \end{cases} \quad (22.88)$$

Then, for $\mathcal{L}(u(\cdot), \mu, v, T)$,

$$\mathcal{L}(u(\cdot), \mu, v, T) := \mu J(u(\cdot), T) + \sum_{l=1}^L v_l g_l(x(T), T) \quad (22.89)$$

it follows that

$$\begin{aligned} 0 &\leq \mathcal{L}(u^\varepsilon(\cdot), \mu, v, T^* - \varepsilon) - \mathcal{L}(u^*(\cdot), \mu, v, T^*) \\ &= \mu h_0(x(T^* - \varepsilon), T^* - \varepsilon) + \sum_{l=1}^L v_l g_l(x(T^* - \varepsilon), T^* - \varepsilon) \\ &\quad - \mu h_0(x^*(T^*), T^*) - \sum_{l=1}^L v_l g_l(x^*(T^*), T^*) \end{aligned}$$

Hence, applying the transversality condition (22.71) we obtain:

$$\begin{aligned} 0 &\leq -\varepsilon \left(\mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \right) + o(\varepsilon) \\ &\quad - \varepsilon \left(\mu \frac{\partial}{\partial x} h_0(x(T^*), T^*) \right. \\ &\quad \left. + \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x(T^*), T^*), f(x(T^*), u^*(T^* - 0), T^*) \right) \\ &= -\varepsilon \left(\mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \right) \\ &\quad + \varepsilon \psi^\top(T^*) f(x(T^*), u^*(T^* - 0), T^*) + o(\varepsilon) \\ &= -\varepsilon \left(\mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \right) \\ &\quad + \varepsilon H(\psi(T^*), x^*(T^*), u^*(T - 0), T^*) + o(\varepsilon) \end{aligned}$$

which, by dividing by ε and tending ε to zero, implies

$$\begin{aligned} & H(\psi(T^*), x^*(T^*), u^*(T-0), T^*) \\ & \geq \mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \end{aligned} \quad (22.90)$$

(b) Analogously, for the needle-shape variation

$$u^\varepsilon(t) := \begin{cases} u^*(t) & \text{if } t \in [0, T^*) \setminus M_\varepsilon \\ u(t) \in \mathcal{U}_{admis} [0, T^*] & \text{if } t \in M_\varepsilon \\ u^*(T^* - 0) & \text{if } t \in [T^*, T^* + \varepsilon] \end{cases} \quad (22.91)$$

it follows that

$$\begin{aligned} 0 & \leq \mathcal{L}(u^\varepsilon(\cdot), \mu, v, T^* + \varepsilon) - \mathcal{L}(u^*(\cdot), \mu, v, T^*) \\ & = \varepsilon \left(\mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \right) \\ & \quad - \varepsilon H(\psi(T^*), x^*(T^*), u^*(T-0), T^*) + o(\varepsilon) \end{aligned}$$

and

$$\begin{aligned} & H(\psi(T^*), x^*(T^*), u^*(T-0), T^*) \\ & \leq \mu \frac{\partial}{\partial T} h_0(x(T^*), T^*) + v_l \frac{\partial}{\partial T} g_l(x(T^*), T^*) \end{aligned} \quad (22.92)$$

Combining (22.88) and (22.91), we obtain (22.87). Theorem is proven. $\square$

Corollary 22.6. (Zero property) *If under the conditions of Theorem 22.13 the functions $h_0(x, T)$, $g_l(x, T)$ ($l = 1, \dots, L$) do not depend on T directly, that is,*

$$\frac{\partial}{\partial T} h_0(x, T) = \frac{\partial}{\partial T} g_l(x, T) = 0 \quad (l = 1, \dots, L)$$

then

$$H(\psi(T^*), x^*(T^*), u^*(T-0), T^*) = 0 \quad (22.93)$$

If, in addition, the stationary case is considered (see (22.83)), then (22.93) holds for all $t \in [0, T^]$, that is,*

$$H(\psi(t), x^*(t), u^*(\psi(t), x^*(t))) = 0 \quad (22.94)$$

Proof. The result directly follows from (22.84) and (22.93). $\square$

22.6.6 Joint optimal control and parametric optimization problem

Consider the nonlinear plant given by

$$\left. \begin{aligned} \dot{x}_a(t) &= f(x_a(t), u(t), t; a), \quad \text{a.e. } t \in [0, T] \\ x_a(0) &= x_0 \end{aligned} \right\} \quad (22.95)$$

at the fixed horizon T , where $a \in \mathbb{R}^p$ is a vector of parameters which also can be selected to optimize the functional (22.44) which in this case is

$$J(u(\cdot), a) = h_0(x_a(T)) \quad (22.96)$$

(A4) It will be supposed that the right-hand side of (22.95) is differentiable on a at all $a \in \mathbb{R}^p$.

In view of this OCP is formulated as follows:

$$\begin{aligned} &\text{Minimize } J(u(\cdot), a) \quad (22.96) \\ &\text{over } u(\cdot) \in \mathcal{U}_{admis}[0, T] \quad \text{and } a \in \mathbb{R}^p \end{aligned} \quad (22.97)$$

Theorem 22.14. (Joint OC and parametric optimization) *If under assumptions (A1)–(A3) and (A4) the pair $(u^*(\cdot), a^*)$ is a solution of the problem (22.85), (22.86) and $x^*(t)$ is the corresponding optimal trajectory, then there exist the vector functions $\psi(t)$ satisfying the system of the adjoint equations (22.68) with $x^*(t)$, $u^*(t)$, a^* and nonnegative constants $\mu \geq 0$ and $v_l \geq 0$ ($l = 1, \dots, L$) such that all four conditions of Theorem 22.11 are fulfilled and, in addition, the following condition to the optimal parameter holds:*

$$\int_{t=0}^T \frac{\partial}{\partial a} H(\psi(t), x^*(t), u^*(t), t; a^*) dt = 0 \quad (22.98)$$

Proof. For this problem $\mathcal{L}(u(\cdot), \mu, v, a)$ is defined by

$$\mathcal{L}(u(\cdot), \mu, v, a) := \mu h_0(x(T)) + \sum_{l=1}^L v_l g_l(x(T)) \quad (22.99)$$

Introduce the matrix $\Delta^a(t) = \frac{\partial}{\partial a} x^*(t) \in \mathbb{R}^{n \times p}$, called the *matrix of sensitivity (with respect to parameter variations)*, which satisfies the following differential equation:

$$\begin{aligned} \dot{\Delta}^a(t) &= \frac{d}{dt} \frac{\partial}{\partial a} x^*(t) = \frac{\partial}{\partial a} \dot{x}^*(t) = \frac{\partial}{\partial a} f(x^*(t), u^*(t), t; a^*) \\ &= \frac{\partial}{\partial a} f(x^*(t), u^*(t), t; a^*) \\ &\quad + \frac{\partial}{\partial x} f(x^*(t), u^*(t), t; a^*) \Delta^a(t), \quad \Delta^a(0) = 0 \end{aligned} \quad (22.100)$$

In view of this and using (22.68), it follows that

$$\begin{aligned}
 0 &\leq \mathcal{L}(u^*(\cdot), \mu, v, a) - \mathcal{L}(u^*(\cdot), \mu, v, a^*) \\
 &= (a - a^*)^\top \Delta^a(T)^\top \left(\mu \frac{\partial}{\partial x} h_0(x^*(T)) + \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x^*(T)) \right) \\
 &\quad + o(\|a - a^*\|) = (a - a^*)^\top \Delta^a(T)^\top \psi(T) + o(\|a - a^*\|) \\
 &= (a - a^*)^\top [\Delta^a(T)^\top \psi(T) - \Delta^a(0)^\top \psi(0)] + o(\|a - a^*\|) \\
 &= (a - a^*)^\top \int_{t=0}^T d[\Delta^a(t)^\top \psi(t)] + o(\|a - a^*\|) \\
 &= (a - a^*)^\top \int_{t=0}^T \left[-\Delta^a(t)^\top \frac{\partial}{\partial x} f(x^*(t), u^*(t), t; a^*)^\top \psi(t) \right. \\
 &\quad \left. + \Delta^a(t)^\top \frac{\partial}{\partial x} f(x^*(t), u^*(t), t; a^*)^\top \psi(t) \right. \\
 &\quad \left. + \frac{\partial}{\partial a} f(x^*(t), u^*(t), t; a^*)^\top \psi(t) \right] dt + o(\|a - a^*\|) \\
 &= (a - a^*)^\top \int_{t=0}^T \frac{\partial}{\partial a} f(x^*(t), u^*(t), t; a^*)^\top \psi(t) dt + o(\|a - a^*\|)
 \end{aligned}$$

But this inequality is possible for any $a \in \mathbb{R}^p$ in a small neighborhood of a^* if and only if the relation (22.98) holds (which may be proved by contradiction). Theorem is proven. $\square$

22.6.7 Sufficient conditions of optimality

The necessary and sufficient conditions of the constrained concave optimization problem on $x \in X \subseteq \mathbb{R}^n$ is (see (21.80))

$$(\partial f(x^*), x - x^*) \leq 0$$

which should be valid for all $x \in X$ (X is supposed to be a convex set and $f(x)$ is concave on X).

Here we will also need an additional assumption concerning the control region.

(A4) The control domain U is supposed to be a *convex body* (i.e., it is convex and has a nonempty interior).

Lemma 22.9. (on a mixed subgradient) Let φ be a convex (or concave) function on $\mathbb{R}^n \times U$ where U is a convex body. Assuming that $\varphi(x, u)$ is differentiable in x and

is continuous in (x, u) , the following inclusion turns out to be valid for any $(x^*, u^*) \in \mathbb{R}^n \times U$:

$$\{(\varphi_x(x^*, u^*), r) \mid r \in \partial_u \varphi(x^*, u^*)\} \subseteq \partial_{x,u} \varphi(x^*, u^*) \quad (22.101)$$

Proof. For any $y \in \mathbb{R}^n$, in view of the convexity of φ and its differentiability on x , it follows that

$$\varphi(x^* + y, u^*) - \varphi(x^*, u^*) \geq (\varphi_x(x^*, u^*), y) \quad (22.102)$$

Similarly, in view of the convexity of φ in u , there exists a vector $r \in \mathbb{R}^r$ such that for any $x^*, y \in \mathbb{R}^n$ and any $\bar{u} \in U$

$$\varphi(x^* + y, u^* + \bar{u}) - \varphi(x^* + y, u^*) \geq (r, \bar{u}) \quad (22.103)$$

So, taking into account the previous inequalities (22.102)–(22.103), we derive

$$\begin{aligned} & \varphi(x^* + y, u^* + \bar{u}) - \varphi(x^*, u^*) \\ &= [\varphi(x^* + y, u^* + \bar{u}) - \varphi(x^* + y, u^*)] \\ &+ [\varphi(x^* + y, u^*) - \varphi(x^*, u^*)] \geq (r, \bar{u}) + (\varphi_x(x^*, u^*), y) \end{aligned} \quad (22.104)$$

Then, by the definition of subgradient (21.69), it means that

$$(\varphi_x(x^*, u^*); r) \subseteq \partial_{x,u} \varphi(x^*, u^*)$$

The concavity case is very similar if we note that $(-\varphi)$ is convex. $\square$

Now we are ready to formulate the central result of this subsection.

Theorem 22.15. (Sufficient condition of optimality) *Let, under assumptions (A1)–(A4), the pair $(x^*(\cdot), u^*(\cdot))$ be an admissible pair and $\psi(t)$ be the corresponding adjoint variable satisfying (22.68). Assume that*

1. $h_0(x)$ and $g_l(x)$ ($l = 1, \dots, L$) are **convex**;
2. $H(\psi(t), x, u, t)$ is **concave** in (x, u) for any fixed $t \in [0, T]$ and any $\psi(t) \in \mathbb{R}^n$.

Then this pair $(x^(\cdot), u^*(\cdot))$ is optimal in the sense of the cost functional $J(u(\cdot)) = h_0(x(T))$ (22.44) if*

$$H(\psi(t), x^*(t), u^*(t), t) = \max_{u \in U} H(\psi(t), x^*(t), u, t) \quad (22.105)$$

at almost all $t \in [0, T]$.

Proof. By (22.105) and in view of the criterion of optimality (21.80), it follows for any $u \in U$ that

$$(\partial_u H(\psi(t), x^*(t), u^*(t), t), u - u^*(t)) \leq 0 \quad (22.106)$$

Then, by concavity of $H(\psi(t), x, u, t)$ in (x, u) , for any admissible pair (x, u) and applying the integration operation, in view of (22.106) we get

$$\begin{aligned}
 & \int_{t=0}^T H(\psi(t), x(t), u, t) dt - \int_{t=0}^T H(\psi(t), x^*(t), u^*(t), t) dt \\
 & \leq \int_{t=0}^T \left[\left(\frac{\partial}{\partial x} H(\psi(t), x^*(t), u^*(t), t), x(t) - x^*(t) \right) \right. \\
 & \quad \left. + (\partial_u H(\psi(t), x^*(t), u^*(t), t), u - u^*(t)) \right] dt \\
 & \leq \int_{t=0}^T \left(\frac{\partial}{\partial x} H(\psi(t), x^*(t), u^*(t), t), x(t) - x^*(t) \right) dt
 \end{aligned} \tag{22.107}$$

Let us introduce the “sensitivity” process $\delta(t) := x(t) - x^*(t)$ which evidently satisfies

$$\begin{aligned}
 \dot{\delta}(t) &= \eta(t) \text{ a.e. } t \in [0, T], \delta(0) = 0 \\
 \eta(t) &:= f(x(t), u(t), t) - f(x^*(t), u^*(t), t)
 \end{aligned} \tag{22.108}$$

Then, in view of (22.68) and (22.107), it follows that

$$\begin{aligned}
 & \frac{\partial}{\partial x} h_0(x^*(T))^\top \delta(T) = -[\psi(T)^\top \delta(T) - \psi(0)^\top \delta(0)] \\
 & = - \int_{t=0}^T d[\psi(t)^\top \delta(t)] = \int_{t=0}^T \frac{\partial}{\partial x} H(\psi(t), x^*(t), u^*(t), t)^\top \delta(t) dt \\
 & \quad - \int_{t=0}^T \psi(t)^\top (f(x(t), u(t), t) - f(x^*(t), u^*(t), t)) dt \\
 & \geq \int_{t=0}^T [H(\psi(t), x(t), u, t) - H(\psi(t), x^*(t), u^*, t)] dt \\
 & \quad - \int_{t=0}^T \psi(t)^\top (f(x(t), u(t), t) - f(x^*(t), u^*(t), t)) dt = 0
 \end{aligned} \tag{22.109}$$

The convexity of $h_0(x)$ and $g_l(x)$ ($l = 1, \dots, L$) and the complementary slackness condition yield $\left(\frac{\partial}{\partial x} g_l(x^*(T)), \delta(T) \right) \geq 0$ and, hence,

$$\begin{aligned}
& \frac{\partial}{\partial x} h_0(x^*(T)) \delta(T) \\
& \leq \left[\frac{\partial}{\partial x} h_0(x^*(T)) + \sum_{l=1}^L v_l \frac{\partial}{\partial x} g_l(x^*(T)) \right]^T \delta(T) \\
& \leq h_0(x(T)) - h_0(x^*(T)) + \sum_{l=1}^L v_l g_l(x^*(T)) \\
& = h_0(x(T)) - h_0(x^*(T))
\end{aligned} \tag{22.110}$$

Combining (22.109) with (22.110), we derive

$$\begin{aligned}
& J(u(\cdot)) - J(u^*(\cdot)) \\
& = h_0(x(T)) - h_0(x^*(T)) \geq \frac{\partial}{\partial x} h_0(x^*(T)) \delta(T) \geq 0
\end{aligned}$$

and, since $u(\cdot)$ is arbitrarily admissible, the desired result follows. $\square$

Remark 22.3. Notice that to check the concavity property of $H(\psi(t), x, u, t)$ (22.70) in (x, u) for any fixed $t \in [0, T]$ and any $\psi(t) \in \mathbb{R}^n$ is not a simple task since it depends on the sign of the $\psi_i(t)$ components. So, the theorem given above may be applied directly practically only for a very narrow class of particular problems where the concavity property may be analytically checked.

Example 22.5. Consider the following variation calculus problem:

$$\begin{aligned}
& \int_{t=0}^T |\dot{x}(t)| dt \rightarrow \inf_{x \in C^1[0, T]} \\
& \dot{x}(t) \geq a > 0, \quad x(0) = 0, \quad x(T) = \xi
\end{aligned} \tag{22.111}$$

It can be represented as an optimal control problem. Indeed, denoting $\dot{x} := u$, the initial problem (22.111) can be represented as

$$\begin{aligned}
& \int_{t=0}^T |u(t)| dt \rightarrow \inf_{u \in C[0, T]} \\
& \dot{x}(t) = u(t), \quad u(t) \geq a > 0, \quad x(0) = 0, \quad x(T) = \xi
\end{aligned} \tag{22.112}$$

According to (22.81), the corresponding Hamiltonian function is

$$H(\psi, x, u, t) := \psi u - \mu |u(t)|$$

where $\psi = \psi(t)$ satisfies the following adjoint ODE (22.68) $\dot{\psi}(t) = 0$ with the transversality condition $\psi(T) = 0$, which gives $\psi(t) = 0$ for all $t \in [0, T]$. The nontriviality condition (22.73) implies $\mu > 0$. So, by the maximality condition (22.80), it follows that

$$\arg \max_{u \geq a} H(\psi, x, u, t) = \arg \min_{u \geq a} \mu |u(t)| = a$$

This leads to the following form for the optimal control: $u(t) = a$. The corresponding optimal curve is $x(t) = at + c$. The boundary conditions imply $c = 0$, $aT = \xi$. So, finally, we may conclude that the initial problem (22.111) has the unique solution $x(t) = at$ if and only if the terminal value $x(T) = \xi > 0$ is equal to $\xi = aT$. In any other cases the solution does not exist.

Example 22.6. Consider the following variation calculus problem

$$\begin{aligned} T &\rightarrow \inf_{x \in C^2[0, T]} \\ |\ddot{x}(t)| &\leq 2 \end{aligned} \quad (22.113)$$

$$x(0) = 1, \quad x(T) = -1, \quad \dot{x}(0) = \dot{x}(T) = 0$$

Let us introduce $u(t) := \ddot{x}(t) \in C[-1, T]$, $x_1(t) := x(t)$ and $x_2(t) := \dot{x}(t)$. Then the initial problem (22.113) can be represented as the following optimal control problem:

$$\begin{aligned} T &\rightarrow \inf_{u \in C[0, T]} \\ \dot{x}_1 &= x_2, \quad \dot{x}_2 = u, \quad |u(t)| \leq 2 \\ x_1(0) &= 1, \quad x_1(T) = -1, \quad x_2(0) = x_2(T) = 0 \end{aligned} \quad (22.114)$$

By (22.81), the corresponding Hamiltonian function is

$$H(\psi, x, u, t) := \psi_1 x_2 + \psi_2 u - \mu, \quad \mu \geq 0$$

since $T = \int_{t=0}^T h \, dt$ with $h = 1$. Here $\psi = \psi(t)$ satisfies the following system of the adjoint ODE (22.68)

$$\dot{\psi}_1(t) = 0, \quad \dot{\psi}_2(t) = -\psi_1$$

which solution is a ramp function

$$\begin{aligned} \psi_1(t) &= c_1 \\ \psi_2(t) &= -c_1 t + c_2, \quad c_1, c_2 = \text{const} \end{aligned}$$

So, by the maximality condition (22.80), it follows that

$$\begin{aligned} u^*(t) &= \arg \max_{|u| \leq 2} H(\psi, x, u, t) \\ &= \arg \max_{|u| \leq 2} H(\psi_2 u) = 2 \operatorname{sign} \psi_2 = 2 \operatorname{sign} (c_2 - c_1 t) \end{aligned}$$

The corresponding optimal curve is

$$\begin{aligned} x_1(t) &= 2 \int_{s=0}^t \int_{\tau=0}^s \operatorname{sign} (c_2 - c_1 \tau) \, d\tau \, ds + c_3 t + c_4 \\ x_2(t) &= \int_{\tau=0}^t \operatorname{sign} (c_2 - c_1 \tau) \, d\tau + c_3 \end{aligned}$$

By the boundary conditions it follows that

$$\begin{aligned} c_4 &= 1, \quad c_3 = 0 \\ -1 &= 2 \int_{s=0}^T \left[\int_{\tau=0}^s \text{sign}(c_2 - c_1 \tau) d\tau \right] ds + 1 \\ 0 &= \int_{\tau=0}^T \text{sign}(c_2 - c_1 \tau) d\tau \end{aligned}$$

Changing the time scale as $\tau' := \tau/T$ leads to

$$0 = \int_{\tau'=0}^1 \text{sign}(c_2 - c'_1 \tau') d\tau', \quad c'_1 = c_1 T$$

which implies $2c_2 = c'_1$. Since $\text{sign}(ab) = \text{sign}(a) \text{sign}(b)$

$$\begin{aligned} -1 &= \int_{s=0}^T \left[\int_{\tau=0}^s \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau \right] ds \\ &= \int_{s=0}^T \left[\int_{\tau=0}^T \chi_{s \geq \tau} \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau \right] ds \\ &= \int_{\tau=0}^T \left(\int_{s=0}^T \chi_{s \geq \tau} ds \right) \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau = \int_{\tau=0}^T (T - \tau) \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau \\ &= \int_{\tau=0}^{T/2} (T - \tau) \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau + \int_{\tau=T/2}^T [T - \tau] \text{sign} \left[c_1 \left(\frac{T}{2} - \tau \right) \right] d\tau \\ &\text{sign } c_1 \left(\int_{\tau=0}^{T/2} (T - \tau) d\tau - \int_{\tau=T/2}^T (T - \tau) d\tau \right) = \frac{T^2}{4} \text{sign } c_1 \end{aligned}$$

This leads to the following conclusion

$$c_1 < 0, \quad T = 2$$

Many other interesting examples can be found in Alexeev *et al.* (1984).

22.7 Dynamic programming

The dynamic programming method is another powerful approach to solving optimal control problems. It provides *sufficient conditions* for testing if some control is optimal or not. The basic idea of this approach consists of considering a family of optimal

control problems with different initial conditions (times and states) and of obtaining some relationships among them via the, so-called, *Hamilton–Jacoby–Bellman equation* (HJB) which is a nonlinear first-order partial differential equation. The optimal control can be designed by maximization (or minimization) of the generalized Hamiltonian involved in this equation. If this HJB equation is solvable (analytically or even numerically) then the corresponding optimal controllers turn out to be given by a nonlinear feedback depending on the optimized plant nonlinearity as well as the solution of the corresponding HJB equation. Such approach actually provides the solutions to the whole family of optimization problems, and, in particular, to the original problem. Such a technique is called “invariant embedding”. The major drawback of the classical HJB method is that it requires that this partial differential equation admits a smooth enough solution. Unfortunately this is not the case even for some very simple situations. To overcome this problem the so-called *viscosity solutions* have been introduced (Crandall & Lions 1983). These solutions are some sort of nonsmooth solutions with a key function to replace the conventional derivatives by set-valued super/sub-differentials maintaining the uniqueness of solutions under very mild conditions. These approaches not only save the DPM as a mathematical method, but make it a powerful tool in tackling optimal control. In this section we do not touch on this approach. But we will discuss the gap between necessary (MP) and sufficient (DPM) conditions.

22.7.1 Bellman’s principle of optimality

Claim 22.1. (Bellman’s principle (BP) of optimality) “Any tail of an optimal trajectory is optimal too.”²

In other words, if some trajectory in the phase space connects the initial $x(0)$ and terminal $x(T)$ points and is optimal in the sense of some cost functional, then the sub-trajectory, connecting any intermediate point $x(t')$ of the same trajectory with the same terminal point $x(T)$, should also be optimal (see Fig. 22.2).

22.7.2 Sufficient conditions for BP fulfilling

Theorem 22.16. (Sufficient condition for BP fulfilling) Let

1. the performance index (a cost functional) $J(u(\cdot))$ with $u(\cdot) \in \mathcal{U}_{admis}[0, T]$ be *separable* for any time $t' \in (0, T)$ such that

$$J(u(\cdot)) = J_1(u_1(\cdot), J_2(u_2(\cdot))) \quad (22.115)$$

where $u_1(\cdot)$ is the control within the time interval $[0, t']$ called the **initial control strategy** and $u_2(\cdot)$ is the control within the time interval $[t', T]$ called the **terminal control strategy**;

² Bellman’s principle of optimality, formulated in Bellman (1960), is as follows: “An optimal policy has the property that whatever the initial state and the initial decisions it must constitute an optimal policy with regards to the state resulting from the first decision.”

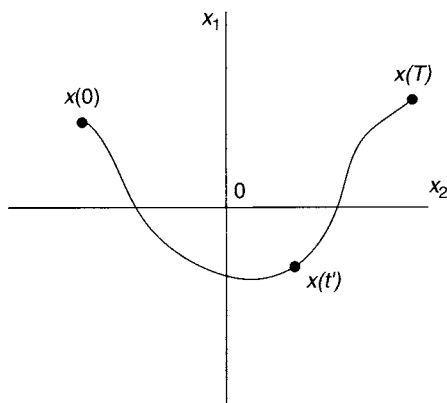


Fig. 22.2. Illustration of Bellman's principle of optimality.

2. the functional $J_1(u_1(\cdot), J_2(u_2(\cdot)))$ is **monotonically nondecreasing with respect to its second argument** $J_2(u_2(\cdot))$, that is,

$$\boxed{\begin{array}{l} J_1(u_1(\cdot), J_2(u_2(\cdot))) \geq J_1(u_1(\cdot), J_2(u'_2(\cdot))) \\ \text{if } J_2(u_2(\cdot)) \geq J_2(u'_2(\cdot)) \end{array}} \quad (22.116)$$

Then Bellman's principle of optimality takes place for this functional.

Proof. For any admissible control strategies $u_1(\cdot)$, $u_2(\cdot)$ the following inequality holds

$$\begin{aligned} J^* &:= \inf_{u \in \mathcal{U}_{\text{admis}}[0, T]} J(u(\cdot)) \\ &= \inf_{u_1 \in \mathcal{U}_{\text{admis}}[0, t'], u_2 \in \mathcal{U}_{\text{admis}}[t', T]} J_1(u_1(\cdot), J_2(u_2(\cdot))) \\ &\leq J_1(u_1(\cdot), J_2(u_2(\cdot))) \end{aligned} \quad (22.117)$$

Select

$$u_2(\cdot) = \arg \inf_{u_2 \in \mathcal{U}_{\text{admis}}[t', T]} J_2(u_2(\cdot)) \quad (22.118)$$

Then (22.117) and (22.118) imply

$$J^* \leq J_1\left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{\text{admis}}[t', T]} J_2(u_2(\cdot))\right) \quad (22.119)$$

So,

$$u_1(\cdot) = \arg \inf_{u_1 \in \mathcal{U}_{\text{admis}}[t', T]} J_1\left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{\text{admis}}[t', T]} J_2(u_2(\cdot))\right) \quad (22.120)$$

leads to

$$J^* \leq \inf_{u_1 \in \mathcal{U}_{\text{admis}}[t', T]} J_1\left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{\text{admis}}[t', T]} J_2(u_2(\cdot))\right) \quad (22.121)$$

Since $J_1(u_1(\cdot), J_2(u_2(\cdot)))$ is monotonically nondecreasing with respect to the second argument, from (22.121) we obtain

$$\begin{aligned} & \inf_{u_1 \in \mathcal{U}_{admis}[t', T]} J_1 \left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{admis}[t', T]} J_2(u_2(\cdot)) \right) \\ & \leq \inf_{u_1 \in \mathcal{U}_{admis}[t', T]} \inf_{u_2 \in \mathcal{U}_{admis}[t', T]} J_1(u_1(\cdot), J_2(u_2(\cdot))) \\ & = \inf_{u \in \mathcal{U}_{admis}[0, T]} J(u(\cdot)) = J^* \end{aligned} \quad (22.122)$$

Combining (22.121) and (22.122), we finally derive that

$$J^* = \inf_{u_1 \in \mathcal{U}_{admis}[t', T]} J_1 \left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{admis}[t', T]} J_2(u_2(\cdot)) \right) \quad (22.123)$$

This proves the desired result. $\square$

Summary 22.3. *In strict mathematical form this fact may be expressed as follows: under the assumptions of the theorem above for any time $t' \in (0, T)$*

$$\begin{aligned} & \inf_{u \in \mathcal{U}_{admis}[0, T]} J(u(\cdot)) \\ & = \inf_{u_1 \in \mathcal{U}_{admis}[t', T]} J_1 \left(u_1(\cdot), \inf_{u_2 \in \mathcal{U}_{admis}[t', T]} J_2(u_2(\cdot)) \right) \end{aligned} \quad (22.124)$$

Corollary 22.7. *For the cost functional*

$$J(u(\cdot)) := h_0(x(T)) + \int_{t=0}^T h(x(t), u(t), t) dt$$

given in the Bolza form (22.41) Bellman's principle holds.

Proof. For any $t' \in (0, T)$ from (22.41) obviously it follows that

$$J(u(\cdot)) = J_1(u_1(\cdot)) + J_2(u_2(\cdot)) \quad (22.125)$$

where

$$\begin{aligned} J_1(u_1(\cdot)) &:= \int_{t=0}^{t'} h(x(t), u_1(t), t) dt \\ J_2(u_2(\cdot)) &:= h_0(x(T)) + \int_{t=t'}^T h(x(t), u_2(t), t) dt \end{aligned} \quad (22.126)$$

The representation (22.125) evidently yields the validity (22.115) and (22.116) for this functional. $\square$

22.7.3 Invariant embedding

22.7.3.1 System description and basic assumptions

Let $(s, y) \in [0, T] \times \mathbb{R}^n$ be “an initial time and state pair” to the following controlled system over $[s, T]$:

$$\left. \begin{aligned} x(t) &= f(x(t), u(t), t), \quad \text{a.e. } t \in [s, T] \\ x(s) &= y \end{aligned} \right\} \quad (22.127)$$

where $x \in \mathbb{R}^n$ is its state vector, and $u \in \mathbb{R}^r$ is the control that may run over a given control region $U \subset \mathbb{R}^r$ with the cost functional in the Bolza form

$$J(s, y; u(\cdot)) := h_0(x(T)) + \int_{t=s}^T h(x(t), u(t), t) dt \quad (22.128)$$

containing the integral term as well as the terminal one and with the *terminal set* $\mathcal{M} \subseteq \mathbb{R}^n$ given by the inequalities (22.42). Here, as before, $u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$. For $s = 0$ and $y = x_0$ this plant coincides with the original one given by (22.40).

Suppose also that assumption (A1) is accepted and, instead of (A2), its small modification holds:

(A2') The maps

$$\left. \begin{aligned} f &: \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R}^n \\ h &: \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R} \\ h_0 &: \mathbb{R}^n \times U \times [0, T] \rightarrow \mathbb{R} \\ g_l &: \mathbb{R}^n \rightarrow \mathbb{R}^n \quad (l = 1, \dots, L) \end{aligned} \right\} \quad (22.129)$$

are **uniformly continuous** in (x, u, t) including t (before in (A2) they were assumed to be only measurable) and there exists a constant L such that for $\varphi = f(x, u, t)$, $h(x, u, t)$, $h_0(x, u, t)$, $g_l(x)$ ($l = 1, \dots, L$) the following inequalities hold:

$$\left. \begin{aligned} \|\varphi(x, u, t) - \varphi(\hat{x}, \hat{u}, t)\| &\leq L \|x - \hat{x}\| \\ \forall t \in [0, T], \quad x, \hat{x} \in \mathbb{R}^n, \quad u \in U \\ \|\varphi(0, u, t)\| &\leq L \quad \forall u, t \in U \times [0, T] \end{aligned} \right\} \quad (22.130)$$

It is evident that under assumptions (A1)–(A2') for any $(s, y) \in [0, T] \times \mathbb{R}^n$ and any $u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$ the optimization problem

$$J(s, y; u(\cdot)) \rightarrow \min_{u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]} \quad (22.131)$$

formulated for the plant (22.127) and for the cost functional $J(s, y; u(\cdot))$ (22.128), admits a unique solution $x(\cdot) := x(\cdot, s, y, u(\cdot))$ and the functional (22.128) is well defined.

Definition 22.8. (The value function) The function $V(s, y)$ defined for any $(s, y) \in [0, T) \times \mathbb{R}^n$ as

$$\left. \begin{aligned} V(s, y) &:= \inf_{u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]} J(s, y; u(\cdot)) \\ V(T, y) &= h_0(y) \end{aligned} \right\} \quad (22.132)$$

is called the **value function** of the optimization problem (22.131).

22.7.3.2 Dynamic programming equation in the integral form

Theorem 22.17. Under assumptions (A1)–(A2') for any $(s, y) \in [0, T) \times \mathbb{R}^n$ the following relation holds

$$\begin{aligned} V(s, y) = \inf_{u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]} & \left\{ \int_{t=s}^{\hat{s}} h(x(t, s, y, u(\cdot)), u(t), t) dt \right. \\ & \left. + V(\hat{s}, x(\hat{s}, s, y, u(\cdot))) \right\} \quad \forall \hat{s} \in [s, T] \end{aligned} \quad (22.133)$$

Proof. The result follows directly from BP of optimality (22.124), but, in view of the great importance of this result, we present the proof again, using the concrete form of the Bolza cost functional (22.128). Denoting the right-hand side of (22.133) by $\bar{V}(s, y)$ and taking into account the definition (22.132), for any $u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$ we have

$$\begin{aligned} V(s, y) &\leq J(s, y; u(\cdot)) \\ &= \int_{t=s}^{\hat{s}} h(x(t, s, y, u(\cdot)), u(t), t) dt + J(\hat{s}, x(\hat{s}); u(\cdot)) \end{aligned}$$

and, taking infimum over $u(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$, it follows that

$$V(s, y) \leq \bar{V}(s, y) \quad (22.134)$$

Hence, for any $\varepsilon > 0$ there exists a control $u_\varepsilon(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$ such that for $x_\varepsilon(\cdot) := x(\cdot, s, y, u_\varepsilon(\cdot))$

$$\begin{aligned} V(s, y) + \varepsilon &\geq J(s, y; u_\varepsilon(\cdot)) \\ &\geq \int_{t=s}^{\hat{s}} h(x(t, s, y, u_\varepsilon(\cdot)), u_\varepsilon(t), t) dt + V(\hat{s}, x_\varepsilon(\hat{s})) \geq \bar{V}(s, y) \end{aligned} \quad (22.135)$$

Tending $\varepsilon \rightarrow 0$ the inequalities (22.134), (22.135) imply the result (22.133) of this theorem. $\square$

Finding a solution $V(s, y)$ to equation (22.133), we would be able to solve the origin optimal control problem putting $s = 0$ and $y = x_0$. Unfortunately, this equation is very difficult to handle because of overcomplicated operations involved on its right-hand side. That's why in the next subsection we will explore this equation further, trying to get another equation for the function $V(s, y)$ with a simpler and more practically used form.

22.7.4 Hamilton–Jacoby–Bellman equation

To simplify the sequent calculations and following Young & Zhou (1999) we will consider the original optimization problem without any terminal set, that is, $\mathcal{M} = \mathbb{R}^n$. This may be expressed with the constraint function equal to

$$g(x) := 0 \cdot \|x\|^2 - \varepsilon \leq 0 \quad (\varepsilon > 0) \quad (22.136)$$

which is true for any $x \in \mathbb{R}^n$. Slater's condition (21.88) is evidently valid (also for any $x \in \mathbb{R}^n$). So, we deal here with the regular case. Denote by $C^1([0, T) \times \mathbb{R}^n)$ the set of all continuously differentiable functions $v : [0, T) \times \mathbb{R}^n \rightarrow \mathbb{R}$.

Theorem 22.18. (The HJB equation) Suppose that under assumptions (A1)–(A2') the value function $V(s, y)$ (22.132) is continuously differentiable, that is, $V \in C^1([0, T) \times \mathbb{R}^n)$. Then $V(s, y)$ is a solution to the following terminal value problem of a first-order partial differential equation, named below the **Hamilton–Jacoby–Bellman (HJB) equation** associated with the original optimization problem (22.131) without terminal set ($\mathcal{M} = \mathbb{R}^n$):

$$\left. \begin{aligned} -\frac{\partial}{\partial t} V(t, x) + \sup_{u \in U} H\left(-\frac{\partial}{\partial x} V(t, x), x(t), u(t), t\right) &= 0 \\ (t, x) \in [0, T) \times \mathbb{R}^n, \quad V(T, x) &= h_0(x), \quad x \in \mathbb{R}^n \end{aligned} \right\} \quad (22.137)$$

where

$$\left. \begin{aligned} H(\psi, x, u, t) &:= \psi^\top f(x, u, t) - h(x(t), u(t), t) \\ t, x, u, \psi &\in [0, T] \times \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R}^n \end{aligned} \right\} \quad (22.138)$$

is the same as in (22.81) with $\mu = 1$ corresponding to the regular optimization problem.

Proof. Fixing $u(t) \equiv u \in U$, by (22.133) with $\hat{s} \downarrow s$ we obtain

$$\begin{aligned} &\frac{V(s, y) - V(\hat{s}, x(\hat{s}, s, y, u(\cdot)))}{\hat{s} - s} \\ &- \frac{1}{\hat{s} - s} \int_{t=s}^{\hat{s}} h(x(t, s, y, u(\cdot)), u(t), t) dt \leq 0 \end{aligned}$$

which implies

$$-\frac{\partial}{\partial t} V(s, y) - \frac{\partial}{\partial x} V(s, y)^\top f(s, y, u) - h(s, u, t) \leq 0$$

resulting in the following inequality

$$0 \geq -\frac{\partial}{\partial t} V(s, y) + \sup_{u \in U} H\left(-\frac{\partial}{\partial x} V(t, x), x(t), u(t), t\right) \quad (22.139)$$

On the other hand, for any $\varepsilon > 0$ and s , closed to $\hat{s}$, there exists a control $u(\cdot) := u_{\varepsilon, \hat{s}}(\cdot) \in \mathcal{U}_{\text{admis}}[s, T]$ for which

$$V(s, y) + \varepsilon(\hat{s} - s) \geq \int_{t=s}^{\hat{s}} h(x(t, s, y, u(\cdot)), u(t), t) dt + V(\hat{s}, x(\hat{s})) \quad (22.140)$$

Since $V \in C^1([0, T) \times \mathbb{R}^n)$, the last inequality leads to the following

$$\begin{aligned} -\varepsilon &\leq -\frac{V(\hat{s}, x(\hat{s})) - V(s, y)}{\hat{s} - s} - \frac{1}{\hat{s} - s} \int_{t=s}^{\hat{s}} h(x(t, s, y, u(\cdot)), u(t), t) dt \\ &= \frac{1}{\hat{s} - s} \int_{t=s}^{\hat{s}} \left[-\frac{\partial}{\partial t} V(t, x(t, s, y, u(\cdot))) \right. \\ &\quad \left. - \frac{\partial}{\partial x} V(t, x(t, s, y, u(\cdot)))^\top f(t, x(t, s, y, u(\cdot)), u) \right. \\ &\quad \left. - h(x(t, s, y, u(\cdot)), u(t), t) \right] dt \\ &= \frac{1}{\hat{s} - s} \int_{t=s}^{\hat{s}} \left[-\frac{\partial}{\partial t} V(t, x(t, s, y, u(\cdot))) \right. \\ &\quad \left. + H\left(-\frac{\partial}{\partial x} V(t, x(t, s, y, u(\cdot))), x(t, s, y, u(\cdot)), u(t), t\right) \right] dt \\ &\leq \frac{1}{\hat{s} - s} \int_{t=s}^{\hat{s}} \left[-\frac{\partial}{\partial t} V(t, x(t, s, y, u(\cdot))) \right. \\ &\quad \left. + \sup_{u \in U} H\left(-\frac{\partial}{\partial x} V(t, x(t, s, y, u(\cdot))), x(t, s, y, u(\cdot)), u(t), t\right) \right] dt \end{aligned} \quad (22.141)$$

which for $\hat{s} \downarrow s$ gives

$$-\varepsilon \leq -\frac{\partial}{\partial t} V(s, y) + \sup_{u \in U} H\left(-\frac{\partial}{\partial x} V(s, y), y, u, s\right) \quad (22.142)$$

Here the uniform continuity property of the functions f and h has been used, namely,

$$\lim_{t \downarrow s} \sup_{y \in \mathbb{R}^n, u \in U} \|\varphi(t, y, u) - \varphi(s, y, u)\| = 0, \quad \varphi = f, h \quad (22.143)$$

Combining (22.139) and (22.142) when $\varepsilon \rightarrow 0$ we obtain (22.137). $\square$

The theorem below, representing the *sufficient conditions of optimality*, is known as the *verification rule*.

Theorem 22.19. (The verification rule) *Accept the following assumptions:*

1. Let $u^*(\cdot) := u^*\left(t, x, \frac{\partial}{\partial x} V(t, x)\right)$ be a solution to the following optimization problem

$$H\left(-\frac{\partial}{\partial x} V(t, x), x, u, t\right) \rightarrow \sup_{u \in U} \quad (22.144)$$

with fixed values x, t and $\frac{\partial}{\partial x} V(t, x)$;

2. Suppose that we can obtain the solution $V(t, x)$ to the HJB equation

$$\left. \begin{aligned} -\frac{\partial}{\partial t} V(t, x) + H\left(-\frac{\partial}{\partial x} V(t, x), x, u^*(\cdot), t\right) &= 0 \\ V(T, x) &= h_0(x), \quad (t, x) \in [0, T) \times \mathbb{R}^n \end{aligned} \right\} \quad (22.145)$$

which for any $(t, x) \in [0, T) \times \mathbb{R}^n$ is unique and smooth, that is, $V \in C^1([0, T) \times \mathbb{R}^n)$;

3. Suppose that for any $(s, x) \in [0, T) \times \mathbb{R}^n$ there exists (a.e. $t \in [s, T]$) a solution $x^*(s, x)$ to the following ODE (ordinary differential equation)

$$\begin{aligned} \dot{x}^*(t) &= f\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right), t\right) \\ x^*(s) &= x \end{aligned} \quad (22.146)$$

Then with $(s, x) = (0, x_0)$ the pair

$$\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right)\right) \quad (22.147)$$

is optimal, that is, $u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right)$ is an **optimal control**.

Proof. The relations (22.138) and (22.145) imply

$$\begin{aligned} \frac{d}{dt} V(t, x^*(t)) &= -\frac{\partial}{\partial t} V(t, x^*(t)) \\ &\quad + \frac{\partial}{\partial x} V(t, x^*(t))^\top f\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right), t\right) \\ &= -h\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right), t\right) \end{aligned} \quad (22.148)$$

Integrating this equality by t within $[s, T]$ leads to the following relation

$$\begin{aligned} V(T, x^*(T)) - V(s, x^*(s)) \\ = - \int_{t=s}^T h\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right), t\right) dt \end{aligned}$$

which, in view of the identity $V(T, x^*(T)) = h_0(x^*(T))$, is equal to the following one

$$V(s, x^*(s)) = h_0(x^*(T)) + \int_{t=s}^T h\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right), t\right) dt \quad (22.149)$$

By (22.133), this last equation means exactly that

$$\left(x^*(t), u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right)\right)$$

is an optimal pair and $u^*\left(t, x^*(t), \frac{\partial}{\partial x} V(t, x^*(t))\right)$ is an optimal control. $\square$

22.8 Linear quadratic optimal control

22.8.1 Nonstationary linear systems and quadratic criterion

Consider in this section the dynamic plants (22.40) in their partial representation when at each time $t \in [0, T]$ the right-hand side of the mathematical model is a linear function with respect to the state vector $x(t)$ and the control action $u(t)$ as well, namely, for almost all $t \in [0, T]$

$$\left. \begin{array}{l} \dot{x}(t) = A(t)x(t) + B(t)u(t) + d(t) \\ x(0) = x_0 \end{array} \right\} \quad (22.150)$$

Here the functional matrices $A(t) \in \mathbb{R}^{n \times n}$ and $B(t) \in \mathbb{R}^{n \times r}$ are supposed to be bounded almost everywhere and the *shifting vector function* $d(t) \in \mathbb{R}^n$ is quadratically integrable, that is,

$$A(\cdot) \in \mathcal{L}^\infty(0, T; \mathbb{R}^{n \times n}), \quad B(\cdot) \in \mathcal{L}^\infty(0, T; \mathbb{R}^{n \times r}), \quad d(\cdot) \in \mathcal{L}^2(0, T; \mathbb{R}^n) \quad (22.151)$$

The admissible control is assumed to be quadratically integrable on $[0, T]$ and the terminal set $\mathcal{M}$ coincides with all space $\mathbb{R}^n$ (no terminal constraints), i.e.,

$$\mathcal{U}_{\text{admis}}[0, T] := \{u(\cdot) : u(\cdot) \in \mathcal{L}^2(0, T; \mathbb{R}^r), \mathcal{M} = \mathbb{R}^n\} \quad (22.152)$$

The cost functional is considered in the form (22.41) with quadratic functions inside, that is,

$$J(u(\cdot)) = \frac{1}{2} x^\top(T) G x(T) + \frac{1}{2} \int_{t=0}^T [x^\top(t) Q(t) x(t) + 2u^\top(t) S(t) x(t) + u^\top(t) R(t) u(t)] dt \quad (22.153)$$

where

$$\begin{aligned} G &\in \mathbb{R}^{n \times n}, \quad Q(\cdot) \in \mathcal{L}^\infty(0, T; \mathbb{R}^{n \times n}) \\ S(\cdot) &\in \mathcal{L}^\infty(0, T; \mathbb{R}^{n \times r}), \quad R(\cdot) \in \mathcal{L}^\infty(0, T; \mathbb{R}^{r \times r}) \end{aligned} \quad (22.154)$$

such that for almost all $t \in [0, T]$

$$G \geq 0, \quad Q(t) \geq 0, \quad R(t) \geq \delta I, \quad \delta > 0 \quad (22.155)$$

Note that all coefficients (except G) in (22.150) and (22.153) are dependent on time t .

22.8.2 Linear quadratic problem

Problem 22.3. (Linear quadratic (LQ) problem) For the dynamic model (22.150) find an admissible control $u^*(\cdot) \in \mathcal{U}_{admis}[0, T]$ such that

$$J(u^*(\cdot)) = \inf_{u(\cdot) \in \mathcal{U}_{admis}[0, T]} J(u(\cdot)) \quad (22.156)$$

where the cost function $J(u(\cdot))$ is given by (22.153).

We will refer to this problem as *the linear quadratic optimal control problem (LQ)*.

22.8.3 Maximum principle for LQ problem

22.8.3.1 MP formulation

Theorem 22.20. (Maximum principle for LQ problem) If a pair $(x^*(t), u^*(\cdot))$ is optimal, then

1. there exists a solution $\psi(t)$ to the following ODE on $[0, T]$

$$\left. \begin{aligned} \dot{\psi}(t) &= -A^\top(t) \psi(t) + Q(t) x^*(t) + S^\top(t) u^*(t) \\ \psi(T) &= -G x^*(T) \end{aligned} \right\} \quad (22.157)$$

2. the optimal control $u^*(\cdot) \in \mathcal{U}_{admis}[0, T]$ is as follows

$$u^*(t) = R^{-1}(t) [B^\top(t) \psi(t) - S(t) x^*(t)] \quad (22.158)$$

Proof. Since in this problem we have no terminal conditions, we deal with the regular case and may take $\mu = 1$. Then by (22.81) and (22.82) it follows that

$$\begin{aligned} H(\psi, x, u, t) &:= \psi^\top [A(t)x + B(t)u + d(t)] \\ &\quad - \frac{1}{2} x^\top(t) Q(t) x(t) - u^\top(t) S(t) x(t) - \frac{1}{2} u^\top(t) R(t) u(t) \end{aligned} \quad (22.159)$$

So,

$$\begin{aligned}\dot{\psi}(t) &= -\frac{\partial}{\partial x} H(\psi(t), x^*(t), u^*(t), t) \\ &= -A^\top(t) \psi(t) + Q(t) x^*(t) + S^\top(t) u^*(t) \\ \psi(T) &= -\frac{\partial}{\partial x} h_0(x^*(T)) = -Gx^*(T)\end{aligned}$$

which proves claim (1) (22.157) of this theorem. Besides, by MP implementation, we have

$$u^*(t) \in \operatorname{Arg\,min}_{u \in \mathbb{R}^r} H(\psi, x^*, u, t)$$

or, equivalently,

$$\frac{\partial}{\partial u} H(\psi, x^*, u^*, t) = B^\top(t) \psi(t) - R(t) u^*(t) - S(t) x^*(t) = 0 \quad (22.160)$$

which leads to claim (2) (22.158). $\square$

22.8.4 Sufficiency condition

Theorem 22.21. (on the sufficiency of MP) *If the control $u^*(t)$ is as in (22.158) and*

$$Q(t) - S(t) R^{-1}(t) S^\top(t) \geq 0 \quad (22.161)$$

then it is a unique optimal one.

Proof. It follows directly from Theorem 22.15 on the sufficient conditions of optimality. The uniqueness is the result of equation (22.160) which has a unique solution if $R(t) \geq \delta I$ a.e. $t \in [0, T]$, $\delta > 0$ (22.155). Besides, the Hessian of the function $H(\psi, x, u, t)$ (22.159) is as follows

$$\left\| \begin{array}{cc} \frac{\partial^2}{\partial x^2} H(\psi, x, u, t) & \frac{\partial^2}{\partial x \partial u} H(\psi, x, u, t) \\ \frac{\partial^2}{\partial u \partial x} H(\psi, x, u, t) & \frac{\partial^2}{\partial u^2} H(\psi, x, u, t) \end{array} \right\| = - \left\| \begin{array}{cc} Q(t) & S(t) \\ S^\top(t) & R(t) \end{array} \right\|$$

Let us show that $\left\| \begin{array}{cc} Q(t) & S(t) \\ S^\top(t) & R(t) \end{array} \right\| \geq 0$. A symmetric block-matrix $\begin{bmatrix} M_{11} & M_{12} \\ M_{12}^\top & M_{22} \end{bmatrix}$ with $M_{22} > 0$ is nonnegative definite, that is,

$$\begin{bmatrix} M_{11} & M_{12} \\ M_{12}^\top & M_{22} \end{bmatrix} \geq 0$$

if and only if

$$M_{11} \geq 0, \quad M_{11} - M_{12} M_{22}^{-1} M_{12}^\top \geq 0$$

So, by the assumption (22.161) of the theorem we have that

$$\left\| \begin{array}{cc} \frac{\partial^2}{\partial x^2} H(\psi, x, u, t) & \frac{\partial^2}{\partial x \partial u} H(\psi, x, u, t) \\ \frac{\partial^2}{\partial u \partial x} H(\psi, x, u, t) & \frac{\partial^2}{\partial u^2} H(\psi, x, u, t) \end{array} \right\| \leq 0$$

This means that the function $H(\psi, x, u, t)$ is concave (not obligatory strictly) on (x, u) for any fixed $\psi(t)$ and any $t \in [0, T]$. $\square$

Corollary 22.8. *If $S(t) \equiv 0$, then the control $u^*(t)$ (22.158) is always uniquely optimal.*

Proof. Under this assumption the inequality (22.161) always holds. $\square$

22.8.5 Riccati differential equation and feedback optimal control

22.8.5.1 Riccati differential equation

Let us introduce the symmetric matrix function $P(t) = P^\top(t) \in C^1(0, T; \mathbb{R}^{n \times n})$ and the vector function $p(t) \in C^1(0, T; \mathbb{R}^n)$ which satisfy (a.e. $t \in [0, T]$) the following ODE:

$$\left. \begin{aligned} -\dot{P}(t) &= P(t) A(t) + A^\top(t) P(t) + Q(t) \\ &\quad - [B^\top(t) P(t) + S(t)]^\top R^{-1}(t) [B^\top(t) P(t) + S(t)] \\ &= P(t) \tilde{A}(t) + \tilde{A}^\top(t) P(t) - P(t) [B(t) R^{-1}(t) B^\top(t)] P(t) + \tilde{Q}(t) \\ P(T) &= G \end{aligned} \right\} \quad (22.162)$$

with

$$\begin{aligned} \tilde{A}(t) &= A(t) - B(t) R^{-1}(t) S(t) \\ \tilde{Q}(t) &= Q(t) - S^\top(t) R^{-1}(t) S(t) \end{aligned} \quad (22.163)$$

and

$$\left. \begin{aligned} -\dot{p}(t) &= \left[(A(t) - B(t) R^{-1}(t) S(t))^\top \right. \\ &\quad \left. - P(t) B(t) R^{-1}(t) B^\top(t) \right] p(t) + P(t) d(t) \\ p(T) &= 0 \end{aligned} \right\} \quad (22.164)$$

Definition 22.9. *We refer to ODE (22.162) as the **Riccati differential equation** and $p(t)$ is referred to as the **shifting vector** associated with the problem (22.156).*

22.8.6 Linear feedback control

Theorem 22.22. (on a linear feedback control) *Assume that*

$$P(t) = P^\top(t) \in C^1(0, T; \mathbb{R}^{n \times n})$$

is a solution of (22.162) and

$$p(t) \in C^1(0, T; \mathbb{R}^n)$$

verifies (22.164). Then the optimal control $u^*(\cdot) \in \mathcal{U}_{\text{admis}}[0, T]$ for the problem (22.156) has the linear feedback form

$$u^*(t) = -R^{-1}(t) [(B^\top(t) P(t) + S(t)) x^*(t) + B^\top(t) p(t)] \quad (22.165)$$

and the optimal cost function $J(u^*(\cdot))$ is as follows

$$\begin{aligned} J(u^*(\cdot)) &= \frac{1}{2} x_0^\top P(0) x_0 + p^\top(0) x_0 \\ &\quad + \frac{1}{2} \int_0^T [2p^\top(t) d(t) - \|R^{-1/2}(t) B^\top(t) p(t)\|^2] dt \end{aligned} \quad (22.166)$$

Proof.

1. Let us try to find the solution of ODE (22.157) in the form

$$\psi(t) = -P(t) x^*(t) - p(t) \quad (22.167)$$

The direct substitution of (22.167) into (22.157) leads to the following identity (t will be suppressed for simplicity):

$$\begin{aligned} (Q - S^\top R^{-1} S) x^* - (A^\top - S^\top R^{-1} B^\top) [-P(t) x^* - p] &= \dot{\psi} \\ &= -\dot{P} x^* - P [A x^* + B(u^*) + d] - \dot{p} = -\dot{P} x^* \\ &\quad - P [A x^* + B R^{-1} [B^\top (-P x^* - p) - S x^*] + d] - \dot{p} \\ &= -\dot{P} x^* - P (A - B R^{-1} [B^\top P + S]) x^* + P B R^{-1} p - P d - \dot{p} \end{aligned}$$

This yields

$$\begin{aligned} 0 &= (\dot{P}(t) + P(t) A(t) + A^\top(t) P(t) + Q(t) \\ &\quad - [B^\top(t) P(t) + S(t)]^\top R^{-1}(t) [B^\top(t) P(t) + S(t)]) x^* \\ &\quad + \dot{p}(t) + [(A(t) - B(t) R^{-1}(t) S(t))]^\top \\ &\quad - P(t) B(t) R^{-1}(t) B^\top(t) p(t) + P(t) d(t) \end{aligned} \quad (22.168)$$

But, in view of (22.162) and (22.164), the right-hand side of (22.168) is identically zero. The transversality condition $\psi(T) = -G x^*(T)$ in (22.162) implies

$$\psi(T) = -P(T) x^*(T) - p(T) = -G x^*(T)$$

which holds for any $x^*(T)$ if $P(T) = G$ and $p(T) = 0$.

2. To prove (22.166) let us apply the chain integration rule for $x^\top(t) P(t) x(t)$ and for $p^\top(t) x(t)$, respectively. In view of (22.150) and (22.162) we obtain

$$\begin{aligned}
 & x^\top(T) P(T) x(T) - x^\top(s) P(s) x(s) \\
 &= x^{*\top}(T) G x^*(T) - x^{*\top}(s) P(s) x^*(s) \\
 &= \int_{t=s}^T \frac{d}{dt} [x^\top(t) P(t) x(t)] dt = \int_{t=s}^T [2x^\top(t) P(t) \dot{x}(t) + x^\top(t) \dot{P}(t) x(t)] dt \\
 &= \int_{t=s}^T \{x^\top(t) ([P(t) B(t) + S^\top(t)] R^{-1}(t) [P(t) B(t) + S^\top(t)]^\top \\
 &\quad - Q(t)) x(t) + 2u^{*\top}(t) B^\top(t) P(t) x(t) + 2d^\top(t) P(t) x(t)\} dt
 \end{aligned} \tag{22.169}$$

and, applying (22.164),

$$\begin{aligned}
 & p^\top(T) x(T) - p^\top(s) x(s) = -p^\top(s) x(s) \\
 &= \int_{t=s}^T \frac{d}{dt} [p^\top(t) x(t)] dt = \int_{t=s}^T [\dot{p}^\top(t) x(t) + p^\top(t) \dot{x}(t)] dt \\
 &= \int_{t=s}^T \{x^\top(t) ([P(t) B(t) + S^\top(t)] R^{-1}(t) B(t) p(t) - P(t) d(t)) \\
 &\quad + p^\top(t) [B(t) u^*(t) + d(t)]\} dt
 \end{aligned} \tag{22.170}$$

Summing (22.169) and (22.170) and denoting

$$\begin{aligned}
 J^*(s, x(s)) &:= \frac{1}{2} x^\top(s) P(s) x(s) \\
 &+ \frac{1}{2} \int_{t=s}^T [x^\top(t) Q(t) x(t) + u^{*\top}(t) R(t) u^*(t) + 2u^{*\top}(t) S(t) x(t)]
 \end{aligned}$$

we get

$$\begin{aligned}
 & J^*(s, x(s)) - \frac{1}{2} x^\top(s) P(s) x(s) - p^\top(s) x(s) \\
 &= \frac{1}{2} \int_{t=s}^T \{u^{*\top}(t) R(t) u^*(t) \\
 &\quad + x^\top(t) [P(t) B(t) + S^\top(t)] R^{-1}(t) [P(t) B(t) + S^\top(t)]^\top x(t) \\
 &\quad + 2x^\top(t) [P(t) B(t) + S^\top(t)]^\top u^*(t) \\
 &\quad + 2x^\top(t) [P(t) B(t) + S^\top(t)]^\top R^{-1}(t) B^\top(t) p(t)
 \end{aligned}$$

$$\begin{aligned}
& + 2u^{*\top}(t) B^\top(t) p(t) + 2p^\top(t) d(t) \} dt \\
& = \frac{1}{2} \int_{t=s}^T \left\{ \|R^{-1/2}(t) [R(t) u^*(t) + [P(t) B(t) + S^\top(t)]^\top x^\top(t)] \right. \\
& \quad \left. + B^\top(t) p(t)\|^2 - \|R^{-1/2}(t) B^\top(t) p(t)\|^2 + 2p^\top(t) d(t) \right\} dt
\end{aligned} \tag{22.171}$$

which, taking $s = 0$, $x(s) = x_0$, and in view of

$$R(t) u^*(t) + [P(t) B(t) + S^\top(t)]^\top x^\top(t) = 0$$

yields (22.166). Theorem is proven. $\square$

Theorem 22.23. (on the uniqueness of the optimal control) *The optimal control $u^*(\cdot) \in \mathcal{U}_{\text{admis}}[0, T]$ is **unique** if and only if the corresponding Riccati differential equation (22.162) has a unique solution $P(t) \geq 0$ on $[0, T]$.*

Proof.

1. *Necessity.* Assume that $u^*(\cdot) \in \mathcal{U}_{\text{admis}}[0, T]$ is unique and is given by (22.165). But this is possible only if $P(t)$ is uniquely defined ($p(t)$ will be uniquely defined automatically). So, the corresponding Riccati differential equation (22.162) should have a unique solution $P(t) \geq 0$ on $[0, T]$.
2. *Sufficiency.* If the corresponding Riccati differential equation (22.162) has a unique solution $P(t) \geq 0$ on $[0, T]$, then, by the previous theorem, $u^*(\cdot)$ is uniquely defined by (22.165) and the dynamics $x^*(t)$ is given by

$$\begin{aligned}
\dot{x}^*(t) &= [A(t) - B(t) R^{-1}(t) (B^\top(t) P(t) + S(t))] x^*(t) \\
&\quad - B(t) R^{-1}(t) B^\top(t) p(t) + d(t)
\end{aligned} \tag{22.172}$$

So, the uniqueness of (22.165) follows from the uniqueness of the solution of ODE (22.172). $\square$

22.8.7 Stationary systems on the infinite horizon

22.8.7.1 Stationary systems and the infinite horizon cost function

Let us consider a stationary linear plant given by the following ODE

$$\left. \begin{aligned} x(t) &= Ax(t) + Bu(t), \quad t \in [0, \infty] \\ x(0) &= x_0, \quad A \in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^{n \times r} \end{aligned} \right\} \tag{22.173}$$

supplied by the quadratic cost function in the Lagrange form, namely,

$$J(u(\cdot)) = \int_{t=0}^{\infty} [x^\top(t) Q x(t) + u^\top(t) R u(t)] dt \tag{22.174}$$

where $0 \leq Q = Q^\top \in \mathbb{R}^{n \times n}$, and $0 < R = R^\top \in \mathbb{R}^{r \times r}$ are the weighting matrices.

The problem is as before: find a control $u^*(\cdot)$ minimizing $J(u(\cdot))$ over all controls within the class of admissible control strategies consisting of all $u(\cdot)$ such that

- the solution of (22.173) exists;
- $u(\cdot) \in L^2(0, \infty; \mathbb{R}^r)$ (otherwise, the criterion (22.174) does not exist).

We will try to solve this problem by two methods: the, so-called, *direct method* and DPM.

22.8.7.2 Direct method

Let us introduce the function $V : \mathbb{R}^n \mapsto \mathbb{R}$ as follows

$$V(x) := x^\top P x \quad (22.175)$$

where the matrix P is a symmetric matrix $P = P^\top \in \mathbb{R}^{n \times n}$. Then, in view of (22.173), we obtain

$$\dot{V}(x(t)) = 2x^\top(t) P \dot{x}(t) = 2x^\top(t) P [Ax(t) + Bu(t)]$$

The integration of this equation leads to the following:

$$\begin{aligned} V(x(T)) - V(x(0)) &= x^\top(T) P x(T) - x_0^\top P x_0 \\ &= \int_{t=0}^T 2x^\top(t) P [Ax(t) + Bu(t)] dt \end{aligned}$$

Adding and subtracting the terms $x^\top(t) Q x(t)$ and $u^\top(t) R u(t)$, the last identity may be rewritten in the following form

$$\begin{aligned} &x^\top(T) P x(T) - x_0^\top P x_0 \\ &= \int_{t=0}^T (2x^\top(t) P [Ax(t) + Bu(t)] + x^\top(t) Q x(t) + u^\top(t) R u(t)) dt \\ &\quad - \int_{t=0}^T [x^\top(t) Q x(t) + u^\top(t) R u(t)] dt = \int_{t=0}^T (x^\top(t) [PA + A^\top P + Q] x(t) \\ &\quad + 2(R^{-1/2} B^\top P x(t))^\top R^{1/2} u(t) + \|R^{1/2} u(t)\|^2) dt \\ &\quad - \int_{t=0}^T [x^\top(t) Q x(t) + u^\top(t) R u(t)] dt \\ &= \int_{t=0}^T (x^\top(t) [PA + AP + Q - PBR^{-1}B^\top P] x(t) \\ &\quad + \|R^{-1/2} B^\top P x(t) + R^{1/2} u(t)\|^2) dt - \int_{t=0}^T [x^\top(t) Q x(t) + u^\top(t) R u(t)] dt \end{aligned}$$

which implies

$$\begin{aligned}
 \int_{t=0}^T [x^T(t) Q x(t) + u^T(t) R u(t)] dt &= x_0^T P x_0 - x^T(T) P x(T) \\
 &+ \int_{t=0}^T \|R^{-1/2} B^T P x(t) + R^{1/2} u(t)\|^2 dt \\
 &+ \int_{t=0}^T x^T(t) [P A + A^T P + Q - P B R^{-1} B^T P] x(t) dt
 \end{aligned} \tag{22.176}$$

Selecting (if it is possible) the matrix P as a solution to the following *matrix Riccati equation*

$$P A + A^T P + Q - P B R^{-1} B^T P = 0 \tag{22.177}$$

from (22.176) we get

$$\begin{aligned}
 \int_{t=0}^T [x^T(t) Q x(t) + u^T(t) R u(t)] dt &= x_0^T P x_0 - x^T(T) P x(T) \\
 &+ \int_{t=0}^T \|R^{-1/2} B^T P x(t) + R^{1/2} u(t)\|^2 dt \\
 &\geq x_0^T P x_0 - x^T(T) P x(T)
 \end{aligned} \tag{22.178}$$

Theorem 22.24. *If for the system (22.173) the pair (A, B) is **stabilizable** and the pair $(Q^{1/2}, A)$ is **observable** then the optimal control $u^*(t)$ minimizing (22.174) is given by*

$$u^*(t) = -R^{-1} B^T P x(t) \tag{22.179}$$

where P is the unique positive definite solution of (22.177) making the closed-loop system asymptotically stable. Moreover, the minimal value of the cost functional is

$$J(u(\cdot)) = \int_{t=0}^{\infty} [x^T(t) Q x(t) + u^T(t) R u(t)] dt = x_0^T P x_0 \tag{22.180}$$

Proof. First, notice that the dynamic system (22.173) closed by an optimal control should be stable. Indeed, suppose that there exists at least one unstable mode of the controlled system. Then, by the observability of the pair (C, A) , it follows that the vector $y(t) = C x(t)$

($Q := C^T C$) should tend to infinity and, hence, $\int_{t=0}^T x^T(t) Q x(t) dt$ tends to infinity with $T \rightarrow \infty$ that cannot correspond to an optimal control. So, the term $x^T(T) P x(T)$ has to tend to zero and can be disregarded. By Theorem 10.8 there exists the unique positive definite solution P of (22.177) which makes the closed-loop system stable since

$$\dot{x}(t) = Ax(t) + Bu^*(t) = (A - BR^{-1}B^T P)x(t) = A_{\text{closed}}x(t)$$

where A_{closed} is defined in (10.26). Taking this P in (22.179) and in view of (22.178) under $u(\cdot) = u^*(\cdot)$ the equality (22.180) holds. Theorem is proven. $\square$

Does the optimal control stabilize nonobservable systems?

Proposition 22.1. *The optimal control does not necessarily stabilize a nonobservable linear stationary system.*

The next simple example illustrates the statement given above.

Example 22.7. *Consider the following linear second-order controllable time invariant system*

$$\dot{x}_1(t) = x_2(t), \quad \dot{x}_2(t) = x_1(t) + u, \quad x_1(0) = x_{10}, \quad x_2(0) = x_{20}$$

Let the cost functional be $J(u(\cdot)) = \int_{t=0}^{\infty} [x^T(t) Q x(t) + u^2(t)] dt$ with $x(t) := \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

and $Q := \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \geq 0$. In our case this system can be represented in the form

(22.173) with $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Notice that this system is unobservable since

the column rank of the observability matrix $\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ is incomplete

and equal to 1. The statement of the proposition becomes evident if we define $y(t) := x_1(t) - x_2(t)$ and represent this system as

$$\begin{aligned} \dot{y}(t) &= -y(t) - u \\ J(u(\cdot)) &= \int_{t=0}^{\infty} [y^2(t) + u^2(t)] dt \end{aligned} \tag{22.181}$$

with

$$\dot{x}_2(t) = y(t) + x_2(t) + u$$

According to Theorem 22.24 the optimal control in (22.181) is $u^*(t) = py(t)$ where $p = \sqrt{2} - 1$. As it is expected, in the optimal system $y(t) \rightarrow 0$ as $t \rightarrow \infty$. Evidently, the second component $x_2(t) \rightarrow \infty$, and, hence, the optimal system is unstable which proves the above proposition. **This effect appears due to instability of the unobservable state component $x_2(t)$!**

The revealed fact permits to indicate the class of stable unobservable optimal systems. If the pair (C, A) is unobservable then there exists a nonsingular linear transformation $Tx = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ such that the system (22.173) can be represented in the, so-called, *canonical observability form*

$$\begin{aligned} \dot{x}_1 &= A_{11} + B_1 u, \quad \dot{x}_2 = A_{21}x_1 + A_{22}x_2 + B_2 u \\ y &= C_1 x_1 \end{aligned} \quad (22.182)$$

with x_1 and x_2 being the observable and unobservable state components. Let us show how this transformation can be found. Select T in the form $T = \begin{pmatrix} v \\ w \end{pmatrix}$ where the matrix $v \in \mathbb{R}^{k \times n}$ consists of k basis row vectors of the observability matrix (9.63)

$$\mathcal{O} := \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

Since the system is unobservable then $k < n$. The matrix $w \in \mathbb{R}^{(n-k) \times n}$ is an arbitrary one such that $\det T \neq 0$. Denote

$$T^{-1} := (N_1 \ N_2), \quad N_1 \in \mathbb{R}^{n \times k}$$

The identity $TT^{-1} = I$ implies

$$I = \begin{pmatrix} v \\ w \end{pmatrix} (N_1 \ N_2) = \begin{pmatrix} vN_1 & vN_2 \\ wN_1 & wN_2 \end{pmatrix}$$

and

$$vN_2 = 0 \quad (22.183)$$

Since vA and vC are in the same basis there exist matrices L_A and L_C such that

$$vA = L_A v, \quad C = L_C v$$

Then

$$\begin{aligned} \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} &= TAT^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + TBu \\ y &= CT^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

In view of (22.183) and the relations

$$\begin{aligned} TAT^{-1} &= \begin{pmatrix} v \\ w \end{pmatrix} A \begin{pmatrix} N_1 & N_2 \end{pmatrix} \\ &= \begin{pmatrix} L_A v \\ wA \end{pmatrix} \begin{pmatrix} N_1 & N_2 \end{pmatrix} = \begin{pmatrix} L_A v N_1 & 0 \\ wA N_1 & wA N_2 \end{pmatrix} \\ CT^{-1} &= \begin{pmatrix} L_C v N_1 & L_C v N_2 \end{pmatrix} = \begin{pmatrix} C N_1 & 0 \end{pmatrix} \end{aligned}$$

we finally get (22.182) and the cost functional (22.174) becomes

$$J(u(\cdot)) = \int_{t=0}^{\infty} [x_1^T(t) Q_1 x_1(t) + u^T(t) R u(t)] dt$$

where $Q_1 = C_1^T C_1$ and the pair (C_1, A_{11}) is observable. Then, by Theorem 22.24, the optimal control is

$$u^*(t) = -R^{-1} B_1^T P_1 x_1(t)$$

with P_1 being the positive-definite solution to the reduced order Riccati equation

$$P_1 A_{11} + A_{11}^T P_1 + Q - P_1 B_1 R^{-1} B_1^T P_1 = 0$$

which makes the first subsystem (with respect to x_1) stable. It is evident that the optimal system (22.173) is stable if the matrix A_{22} is Hurwitz (stable). According to the PBH test 9.1, such systems are called *detectable*. Finally we may formulate the following claim.

Claim 22.2. *The linear time invariant system (22.173) optimal in the sense of the cost functional (22.174) is stable if and only if this system is **stabilizable** and **detectable**.*

22.8.7.3 DPM approach

Consider the following HBJ equation (22.137):

$$\left. \begin{aligned} -\bar{h} + \sup_{u \in U} H\left(-\frac{\partial}{\partial x} V(x), x, u\right) &= 0 \\ x \in \mathbb{R}^n, \bar{h} = \text{const}, V(0) &= 0 \end{aligned} \right\} \quad (22.184a)$$

with

$$\begin{aligned} H(\psi, x, u) &:= \psi^T (Ax + Bu) - x^T Qx - u^T Ru \\ x, u, \psi &\in [0, \infty] \times \mathbb{R}^n \times \mathbb{R}^r \times \mathbb{R}^n \end{aligned} \quad (22.185)$$

Theorem 22.25. (Verification rule for LQ-problem) *If the control u^* is a maximizing vector for (22.185) with some $\bar{h} = \text{const}$, that is,*

$$u^* = -\frac{1}{2} R^{-1} B^T \frac{\partial}{\partial x} V(x)$$

where $V(x)$ is a solution to the following HJ equation:

$$-\frac{\partial}{\partial x} V^\top(x) Ax - x^\top Qx + \frac{1}{4} \frac{\partial}{\partial x} V^\top(x) BR^{-1} B^\top \frac{\partial}{\partial x} V(x) = \bar{h}$$

and the closed-loop system is stable, then such u^* is an optimal control.

Proof. It is evident that only admissible control may be stabilizing (if not, the cost function does not exist). By (22.185) for any stabilizing $u(\cdot)$ it follows that

$$H\left(-\frac{\partial}{\partial x} V(x^*), x^*, u^*\right) = \bar{h}, H\left(-\frac{\partial}{\partial x} V(x), x, u\right) \leq \bar{h}$$

and, hence,

$$H\left(-\frac{\partial}{\partial x} V(x), x, u\right) \leq H\left(-\frac{\partial}{\partial x} V(x^*), x^*, u^*\right)$$

which, after integration, leads to the following inequality

$$\begin{aligned} & \int_{t=0}^{\infty} \left[-\frac{\partial}{\partial x} V^\top(x) (Ax + Bu) - x^\top Qx - u^\top Ru \right] dt \\ & \leq \int_{t=0}^{\infty} \left[-\frac{\partial}{\partial x} V^\top(x^*) (Ax^* + Bu^*) - x^{*\top} Qx^* - u^{*\top} Ru^* \right] dt \end{aligned}$$

or, equivalently,

$$\begin{aligned} & \int_{t=0}^T [x^{*\top} Qx^* + u^{*\top} Ru^*] dt \leq \int_{t=0}^T [x^\top Qx + u^\top Ru] dt + \int_{t=0}^{\infty} d(V(x) - V(x^*)) \\ & \quad V(x(0)) = V^\top(x^*(0)) \int_{t=0}^T [x^\top Qx + u^\top Ru] dt + V(x(T)) - V^\top(x^*(T)) \end{aligned}$$

Within the class of stabilizing strategies we have

$$V(x(T)) - V^\top(x^*(T)) \xrightarrow{T \rightarrow \infty} 0$$

which, in view of the last inequality, shows that $u^*(\cdot)$ is an optimal control. $\square$

Try to find the solution to (22.185) as $V(x) = x^\top Px$ with $P = P^\top \geq 0$. This implies $\frac{\partial}{\partial x} V(x) = 2Px$, and, hence,

$$\begin{aligned} & -2x^\top P(Ax - BR^{-1}B^\top Px) - x^\top Qx - x^\top PBR^{-1}B^\top Px \\ & = x^\top (-PA - A^\top P - Q + PBR^{-1}B^\top P)x = 0 \end{aligned}$$

The last equation is identically fulfilled for any $x \in \mathbb{R}^n$ if P is the solution to the same Riccati matrix equation as in (22.177) for a stabilizable and observable system. So, finally, the optimal control is $u^*(t) = -R^{-1}B^\top Px(t)$ which naturally coincides with (22.179).

22.9 Linear-time optimization

22.9.1 General result

For this problem the *cost functional* is

$$\boxed{J(u(\cdot)) = T} \quad (22.186)$$

It can be obtained from the Bolza form functional

$$J(u(\cdot)) := h_0(x(T)) + \int_{t=0}^T h(x(t), u(t), t) dt$$

if we put $h_0(x) \equiv 0$, $h(x, u, t) \equiv 1$. Then for a linear plant, given by (22.150), the Hamiltonian (22.81) is

$$H(\psi, x, u, t) := \psi^\top [A(t)x + B(t)u + d(t)] - \mu \quad (22.187)$$

and, hence, the maximality condition (22.69) becomes as follows:

$$\begin{aligned} u^*(t) &\in \operatorname{Argmax}_{u \in U} \psi^\top(t) [A(t)x(t) + B(t)u(t) + d(t)] \\ &= \operatorname{Argmax}_{u \in U} \psi^\top(t) B(t)u(t) = \operatorname{Argmax}_{u \in U} [B^\top(t)\psi(t)]^\top u(t) \\ &= \operatorname{Argmax}_{u \in U} \sum_{k=1}^r [B^\top(t)\psi(t)]_k u_k(t) \end{aligned} \quad (22.188)$$

Theorem 22.26. (on linear time-optimal control) *If the set U of the admissible control values is a polytope defined by*

$$U := \{u \in \mathbb{R}^r : u_k^- \leq u_k(t) \leq u_k^+, k = 1, \dots, r\} \quad (22.189)$$

then the optimal control (22.188) is as follows

$$u_k^*(t) = \begin{cases} u_k^+ & \text{if } [B^\top(t)\psi(t)]_k > 0 \\ u_k^- & \text{if } [B^\top(t)\psi(t)]_k < 0 \\ \text{any } \bar{u} \in U & \text{if } [B^\top(t)\psi(t)]_k = 0 \end{cases} \quad (22.190)$$

and it is unique.

Proof. Formula (22.190) follows directly from (22.188), (22.189) and the uniqueness is the consequence of the theorem on the sufficient condition of the optimality which demands the concavity (and not obligatory strict) of the Hamiltonian with respect to (x, u) for any fixed ψ , which is evidently fulfilled for the Hamiltonian function (22.187) which is linear on x and u . $\square$

22.9.2 Theorem on n -intervals for stationary linear systems

Consider in detail the partial case of linear systems (22.150) when the matrices of the system are constant, that is, $A(t) = A$, $B(t) = B$. For this case the result below has been obtained in Feldbaum (1953) and is known as the *theorem on n -intervals*. But first, let us prove an axillary lemma.

Lemma 22.10. *If $\lambda_1, \lambda_2, \dots, \lambda_m$ are real numbers and $f_1(t), \dots, f_m(t)$ are the polynomials with real coefficients and having the orders $k_1, \dots, k_m$, correspondingly. Then the function*

$$\varphi(t) = \sum_{i=1}^m f_i(t) e^{\lambda_i t} \quad (22.191)$$

has a number of real roots which does not exceed

$$n_0 := k_1 + \dots + k_m + m - 1 \quad (22.192)$$

Proof. To prove this result let us use the induction method.

1. For $m = 1$ the lemma is true. Indeed, in this case the function $\varphi(t) = f_1(t) e^{\lambda_1 t}$ has the number of roots coinciding with k_1 since $e^{\lambda_1 t} > 0$ for any t .
2. Suppose that this lemma is valid for $m - 1 > 0$. Then let us prove that it holds for m . Multiplying (22.191) by $e^{-\lambda_m t}$ we obtain

$$\varphi(t) e^{-\lambda_m t} = \sum_{i=1}^{m-1} f_i(t) e^{(\lambda_i - \lambda_m)t} + f_m(t) \quad (22.193)$$

Differentiation by t the relation (22.193) $(k_m + 1)$ -times implies

$$\frac{d^{(k_m+1)}}{dt^{(k_m+1)}} (\varphi(t) e^{-\lambda_m t}) = \sum_{i=1}^{m-1} \tilde{f}_i(t) e^{(\lambda_i - \lambda_m)t} := \varphi_{k_m+1}(t)$$

where $\tilde{f}_i(t)$ are the polynomials of the same order as $f_i(t)$. By the supposition before, the function $\varphi_1(t)$ has a number of roots which do not exceed

$$n_{k_m+1} := k_1 + \dots + k_{m-1} + m - 2$$

Since between two roots of continuously differentiable function, there is at least one root of its derivative, then the function $\varphi_{k_m}(t) := \frac{d^{k_m}}{dt^{k_m}} (\varphi(t) e^{-\lambda_m t})$ will have $n_{k_m} = n_{k_m+1} + 1$. Continuing this process, finally we get that the function $\varphi_0(t) := \varphi(t) e^{-\lambda_m t}$ will have

$$\begin{aligned} n_0 &= n_1 + 1 = n_2 + 2 = \dots = n_{k_m+1} + (k_m + 1) \\ &= (k_1 + \dots + k_{m-1} + m - 2) + (k_m + 1) = k_1 + \dots + k_m + m - 1 \end{aligned}$$

And, since $e^{-\lambda_m t} > 0$ always, we may conclude that $\varphi(t)$ has the same number of roots as $\varphi_0(t)$. Lemma is proven. $\square$

Now we are ready to prove the main result of this section.

Theorem 22.27. (Feldbaum 1953) *If the matrix $A \in \mathbb{R}^{n \times n}$ has only real eigenvalues, then the number of switches of any component of the optimal control (22.190) does not exceed $(n - 1)$, that is, a number of the intervals, where each component of the optimal program (22.190) is constant, does not exceed n .*

Proof. Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be the different eigenvalues of the matrix A and $r_1, r_2, \dots, r_m$ are their multiplicity numbers, correspondingly. Then a general solution of the adjoint system of equations $\dot{\psi}(t) = -A^\top \psi(t)$ may be represented as

$$\psi_i(t) = \sum_{j=1}^m p_{ij}(t) e^{-\lambda_j t}, \quad i = 1, \dots, n \quad (22.194)$$

where $p_{ij}(t)$ are polynomials on t whose order does not exceed $(r_j - 1)$. Substituting (22.194) into (22.188) implies

$$\begin{aligned} u_k^*(t) &= \frac{u_k^+}{2} \left[1 + \operatorname{sign} \left(\sum_{i=1}^n b_{ik} \psi_i(t) \right) \right] + \frac{u_k^-}{2} \left[1 - \operatorname{sign} \left(\sum_{i=1}^n b_{ik} \psi_i(t) \right) \right] \\ &= \frac{u_k^+}{2} \left[1 + \operatorname{sign} \left(\sum_{i=1}^n b_{ik} \sum_{j=1}^m p_{ij}(t) e^{-\lambda_j t} \right) \right] \\ &\quad + \frac{u_k^-}{2} \left[1 - \operatorname{sign} \left(\sum_{i=1}^n b_{ik} \sum_{j=1}^m p_{ij}(t) e^{-\lambda_j t} \right) \right] \\ &= \frac{u_k^+}{2} \left[1 + \operatorname{sign} \left(\sum_{j=1}^m \tilde{p}_{kj}(t) e^{-\lambda_j t} \right) \right] + \frac{u_k^-}{2} \left[1 - \operatorname{sign} \left(\sum_{j=1}^m \tilde{p}_{kj}(t) e^{-\lambda_j t} \right) \right] \end{aligned}$$

where $\tilde{p}_{kj}(t)$ are the polynomials on t , whose order does not exceed $(r_j - 1)$, equal to

$$\tilde{p}_{kj}(t) := \sum_{i=1}^n b_{ik} p_{ij}(t) \quad (22.195)$$

Now the number of switches is defined by the number of the roots of the polynomials (22.195). Applying directly the lemma above, we obtain that the polynomial function $\sum_{j=1}^m \tilde{p}_{kj}(t) e^{-\lambda_j t}$ has a number of real roots which do not exceed

$$(r_1 - 1) + (r_2 - 1) + \dots + (r_k - 1) + (k - 1) = r_1 + \dots + r_k - 1 = n - 1$$

Theorem is proven. $\square$